

BFS Traversal

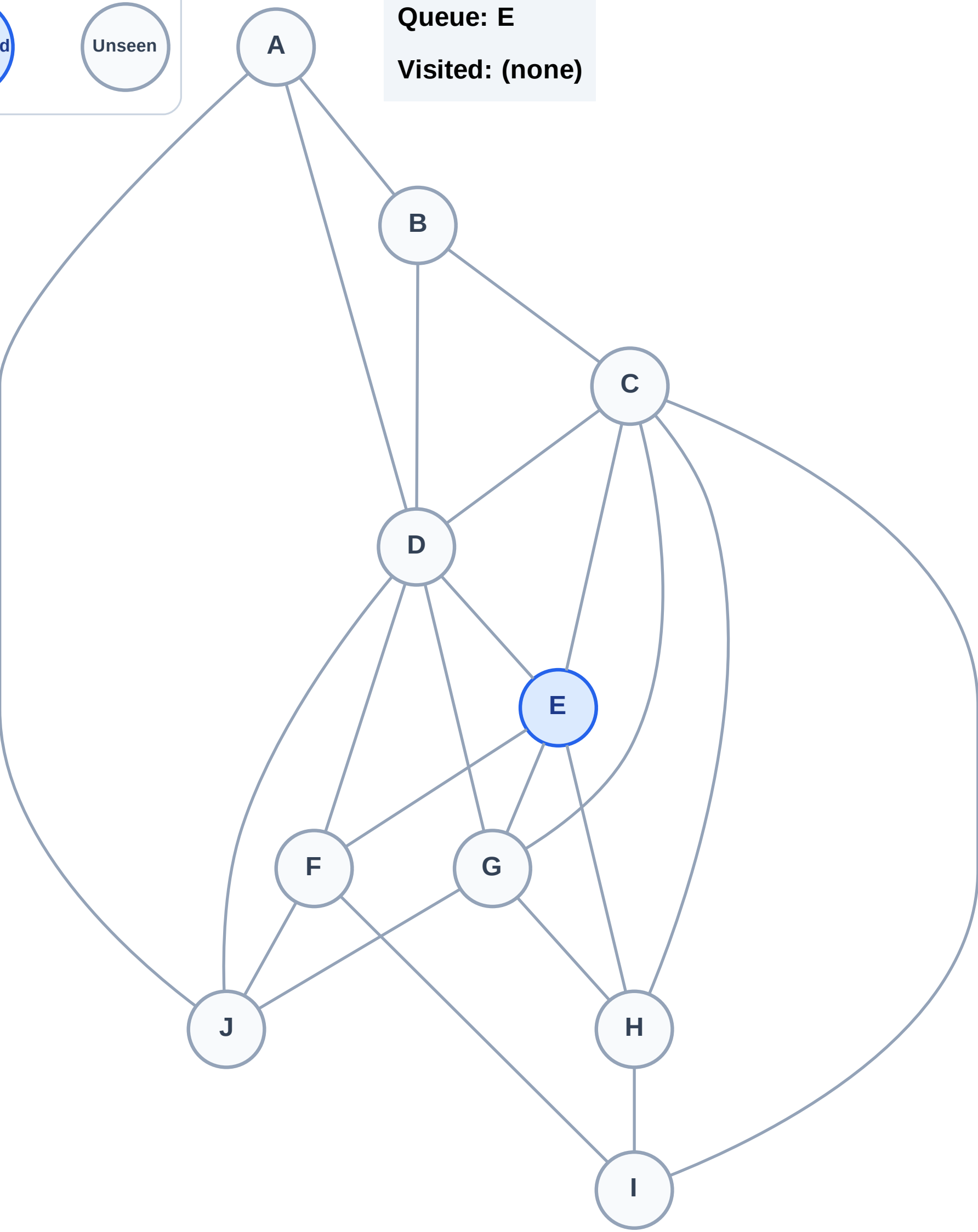
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

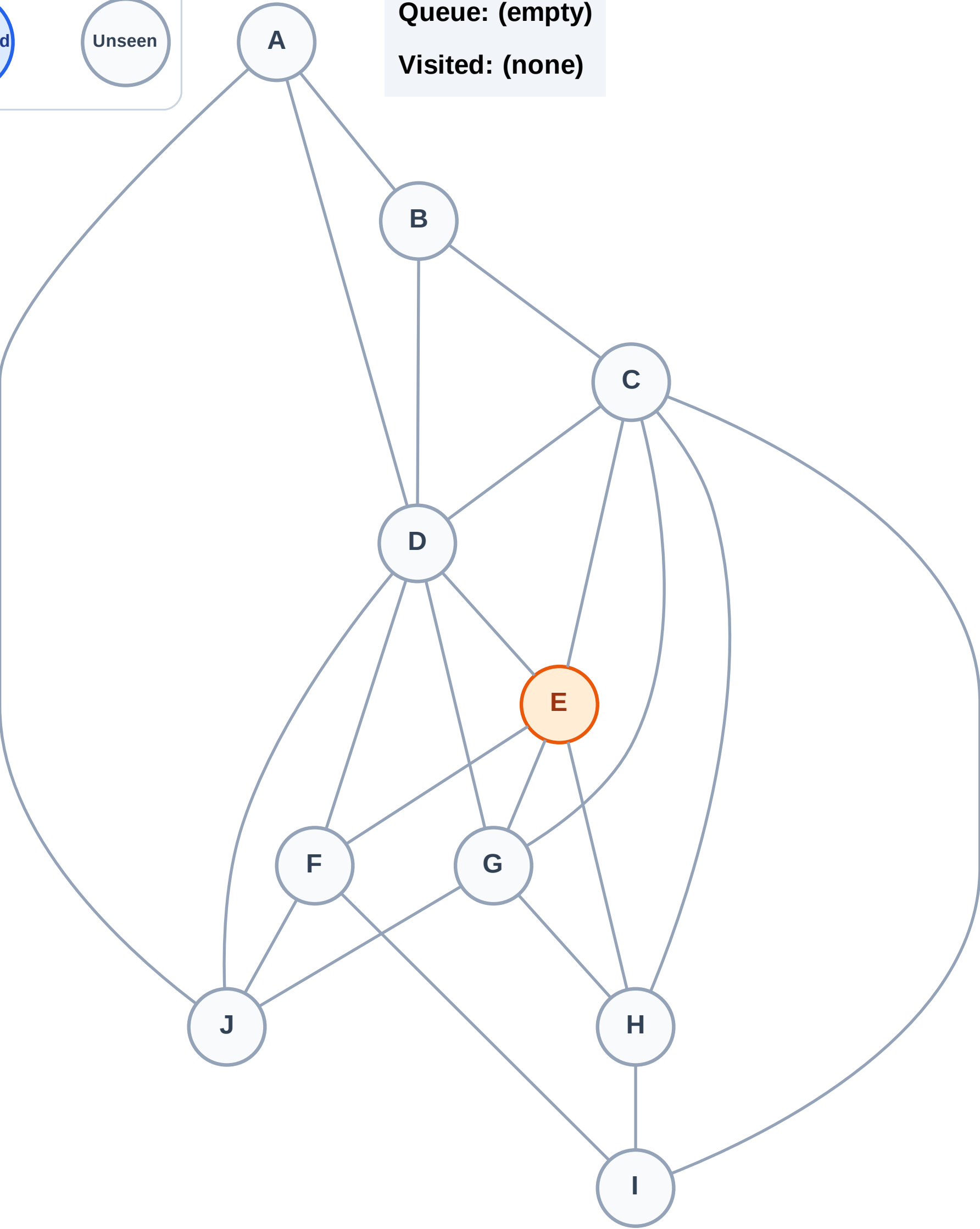
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

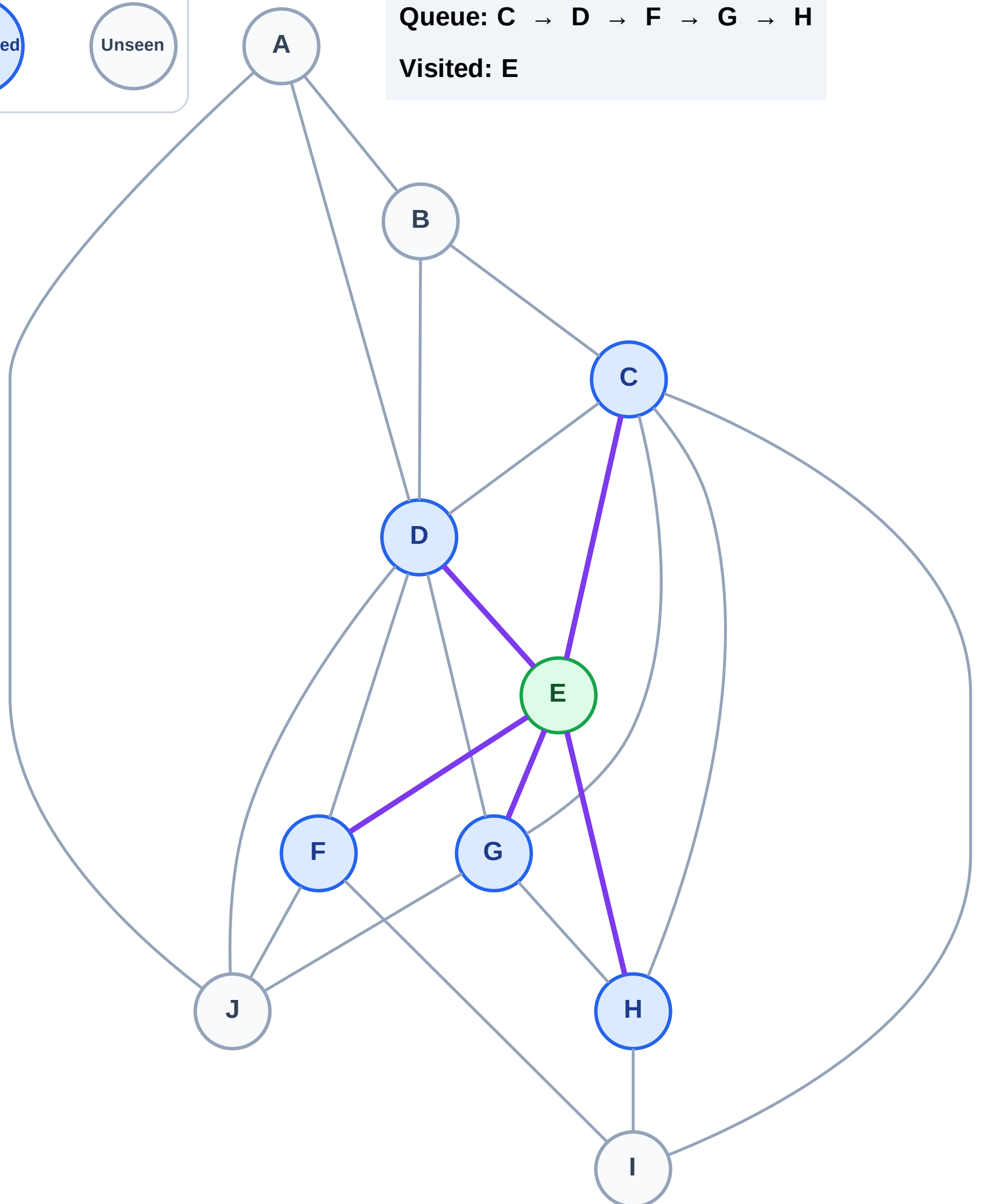
Step 3: Discover C, D, F, G, H from E

Legend



Queue: C → D → F → G → H

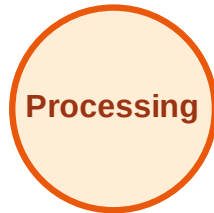
Visited: E



BFS Traversal

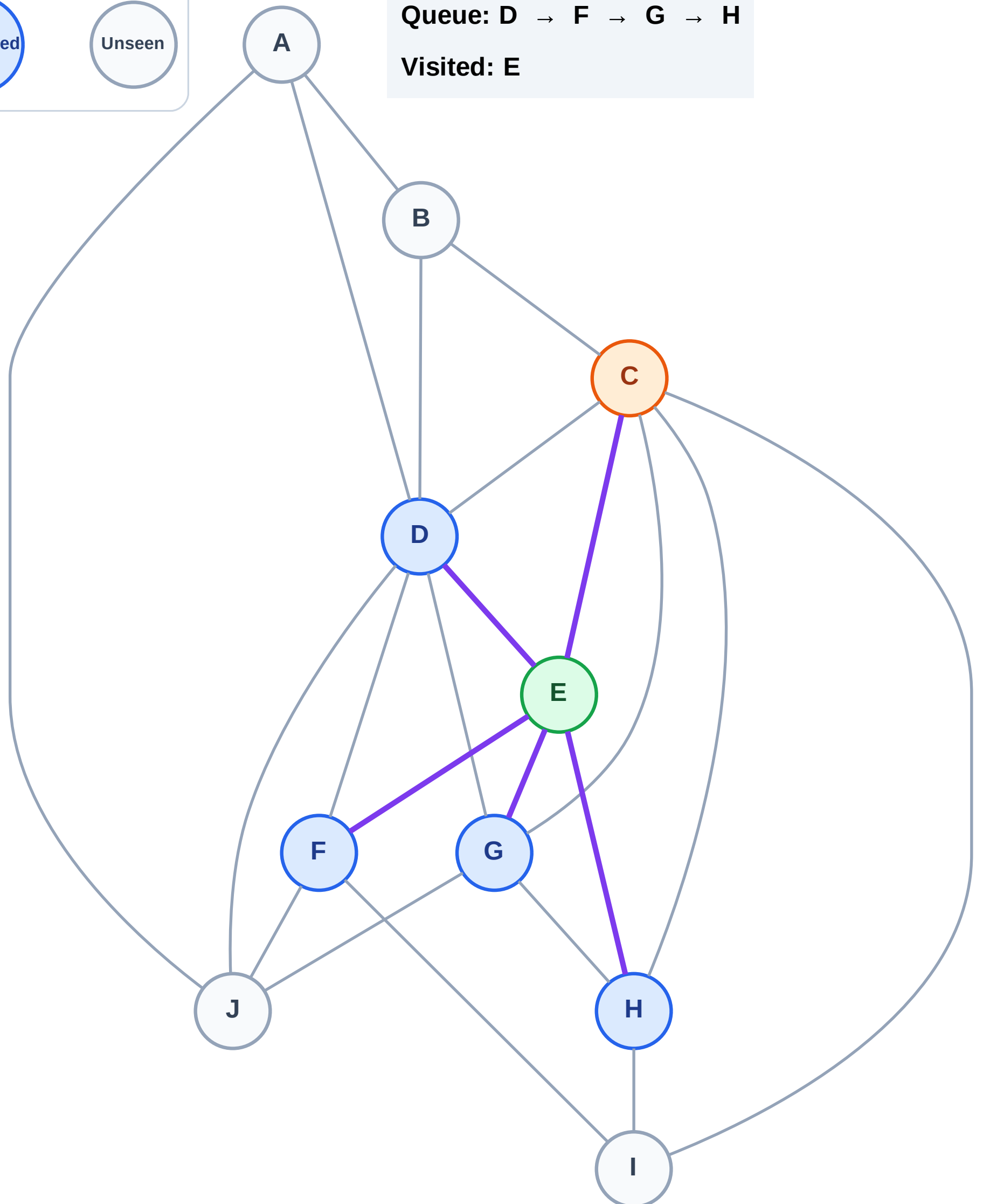
Step 4: Process node C

Legend



Queue: D → F → G → H

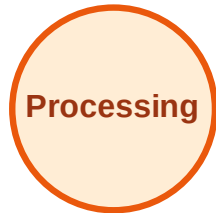
Visited: E



BFS Traversal

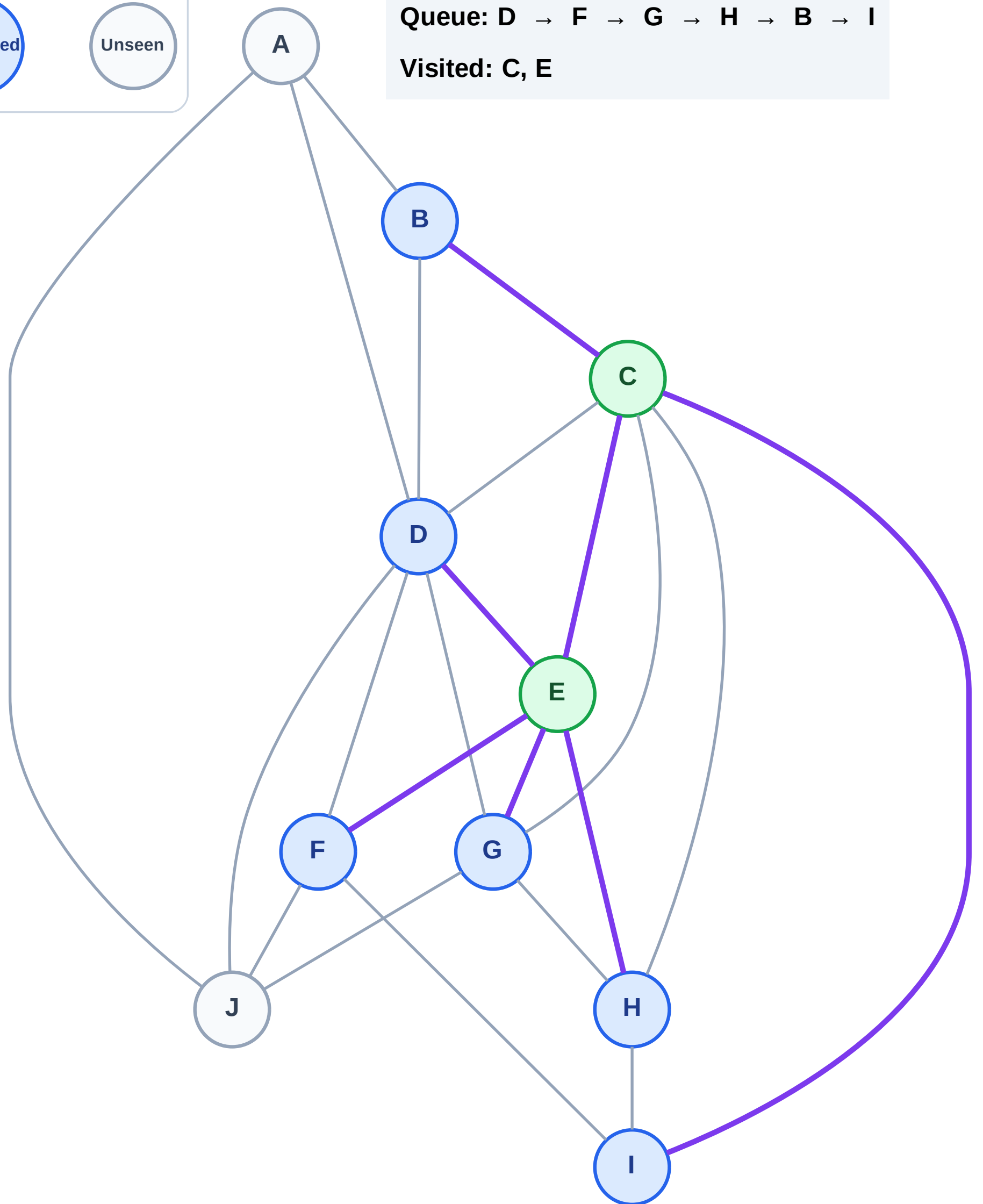
Step 5: Discover B, I from C

Legend



Queue: D → F → G → H → B → I

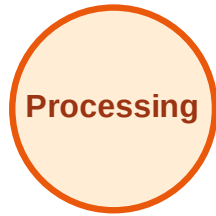
Visited: C, E



BFS Traversal

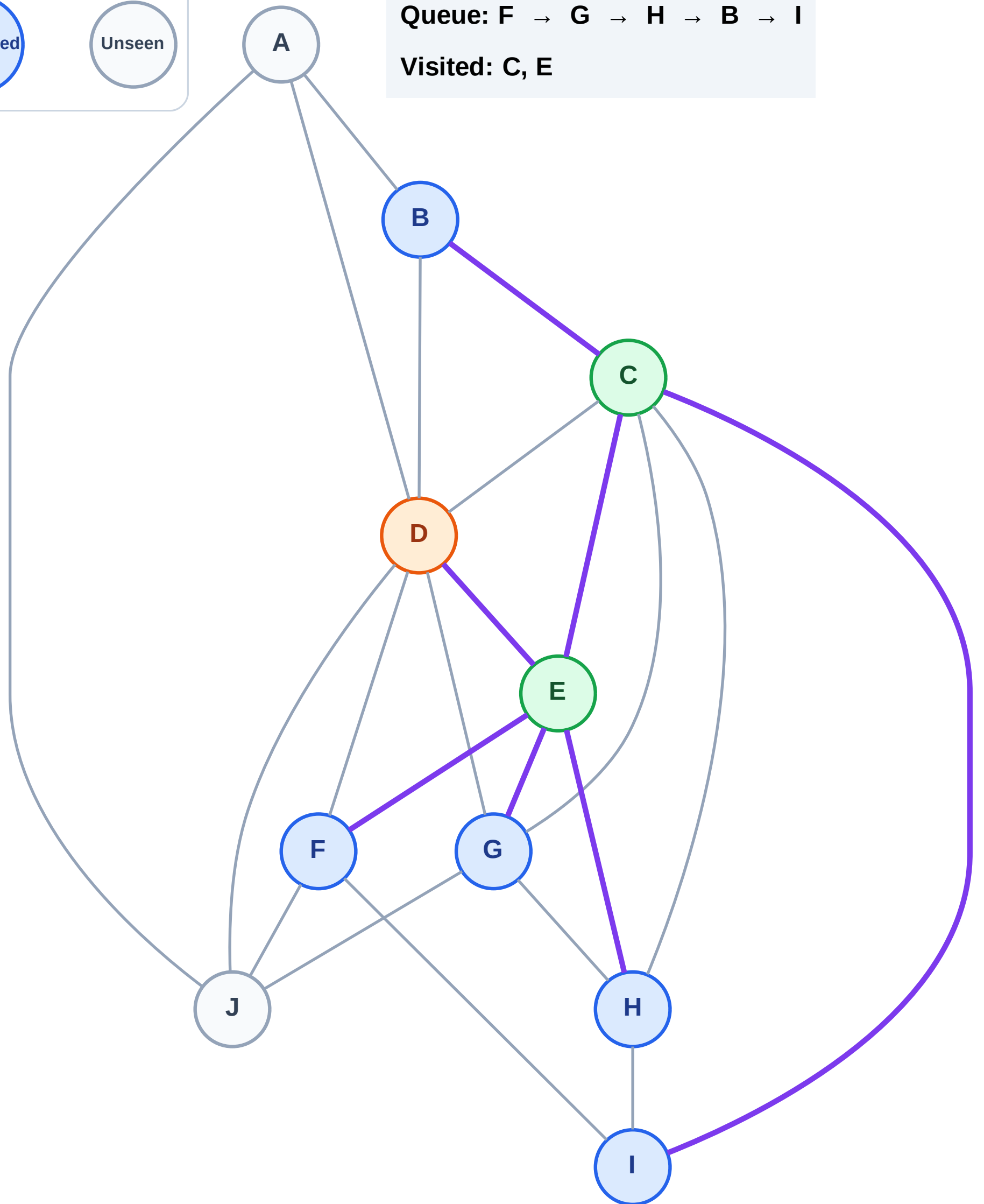
Step 6: Process node D

Legend



Queue: F → G → H → B → I

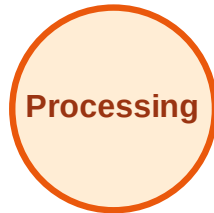
Visited: C, E



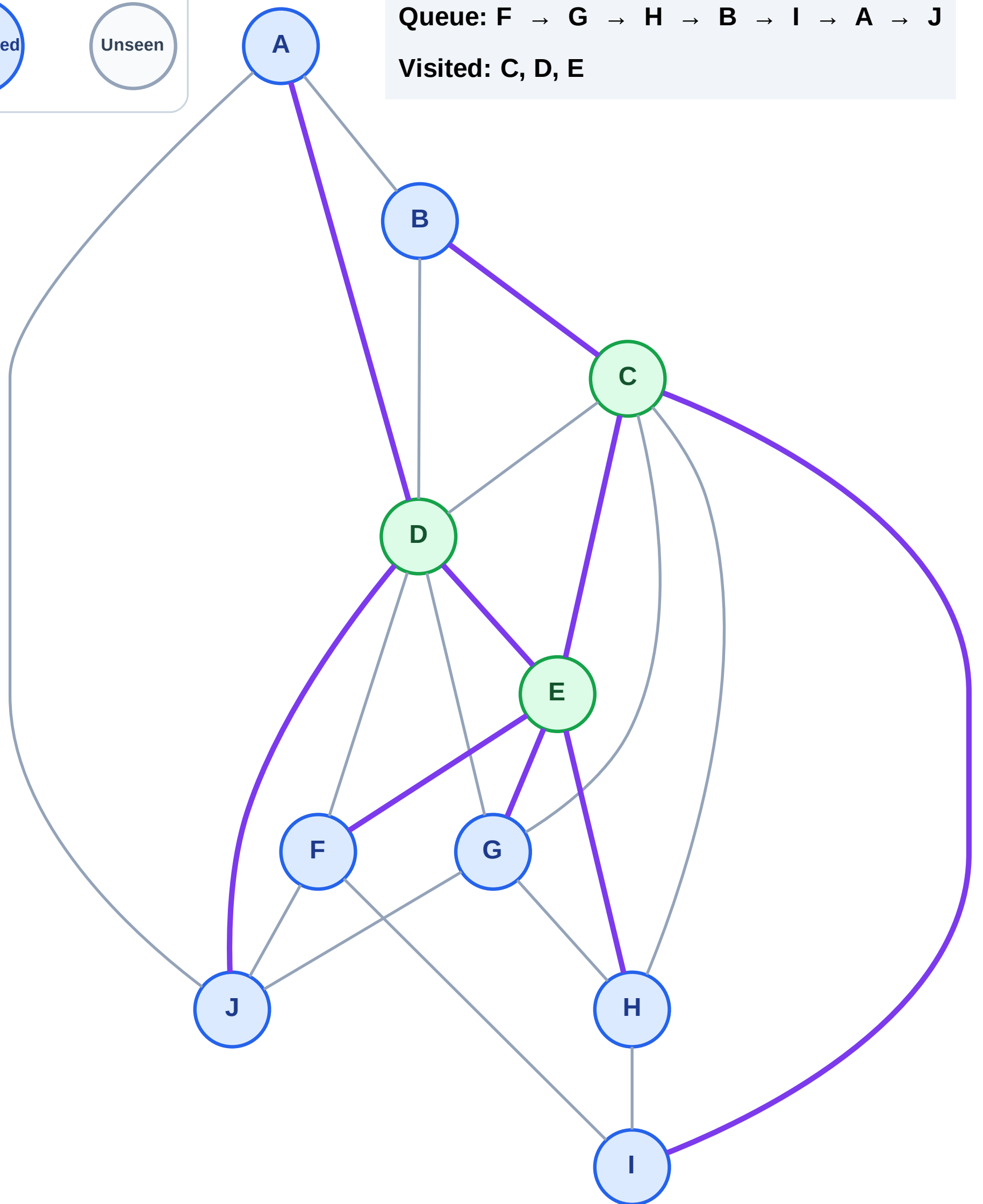
BFS Traversal

Step 7: Discover A, J from D

Legend



Queue: F → G → H → B → I → A → J
Visited: C, D, E



BFS Traversal

Step 8: Process node F

Legend

Complete

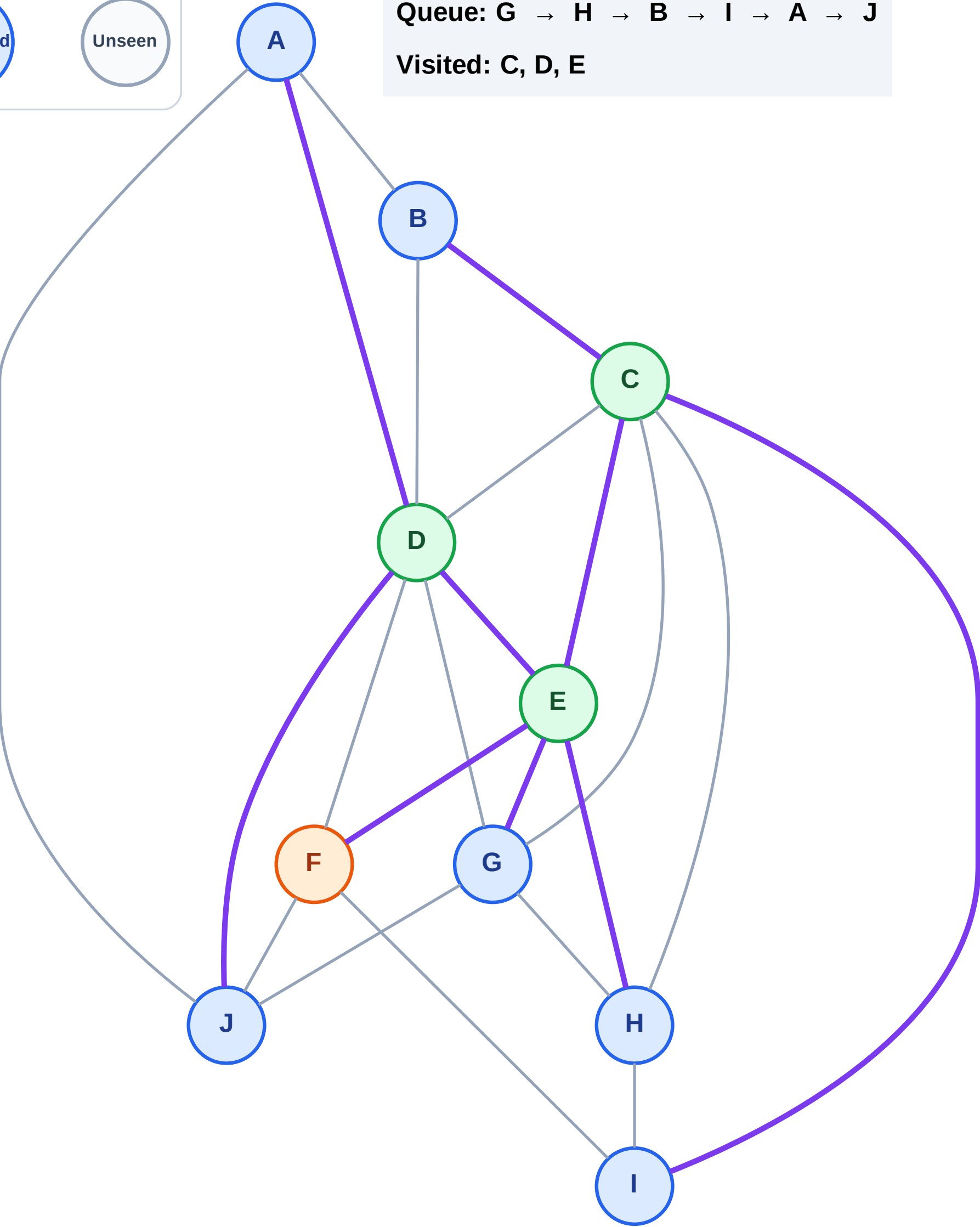
Processing

Discovered

Unseen

Queue: G → H → B → I → A → J

Visited: C, D, E



BFS Traversal

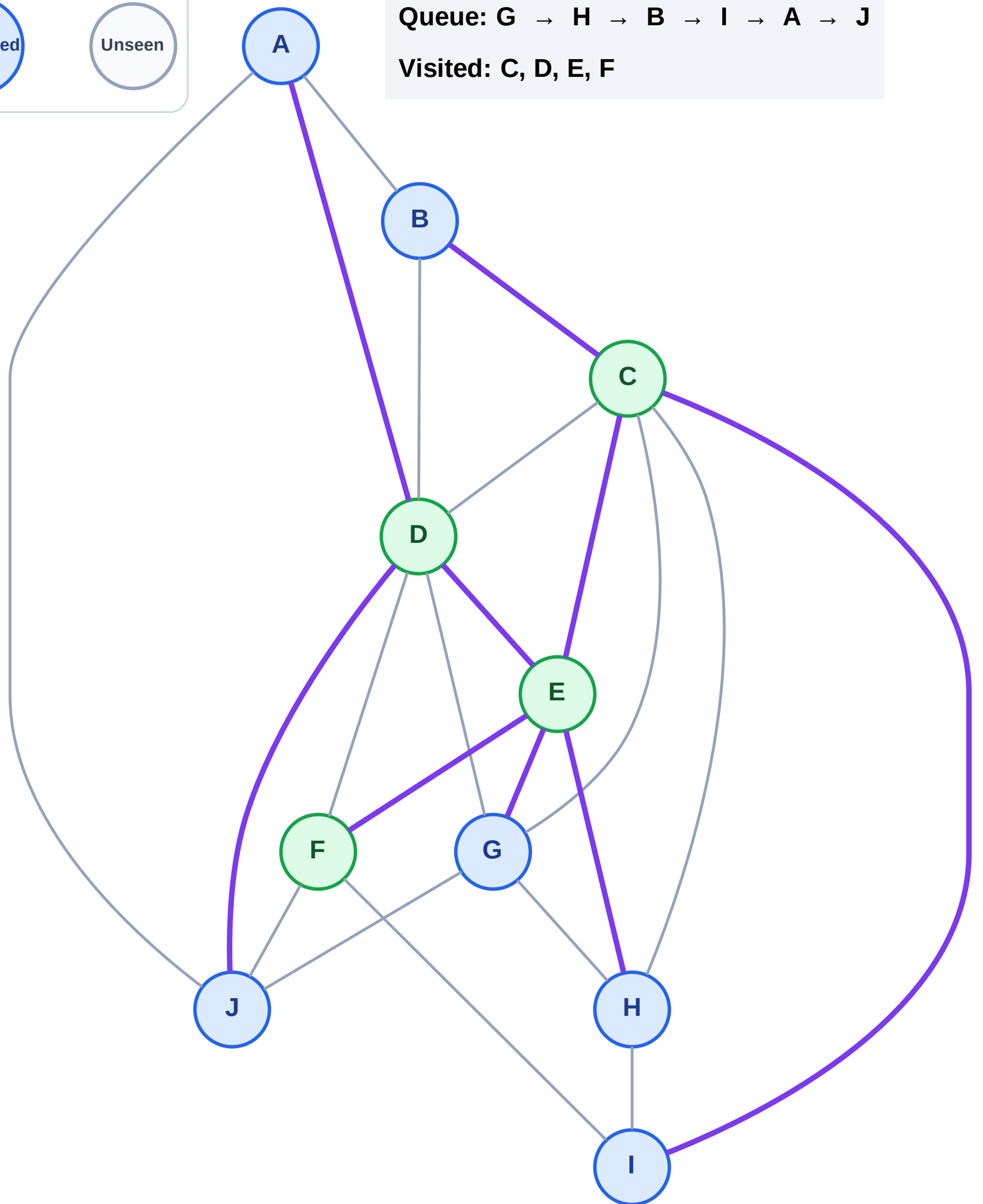
Step 9: No new neighbors from F

Legend

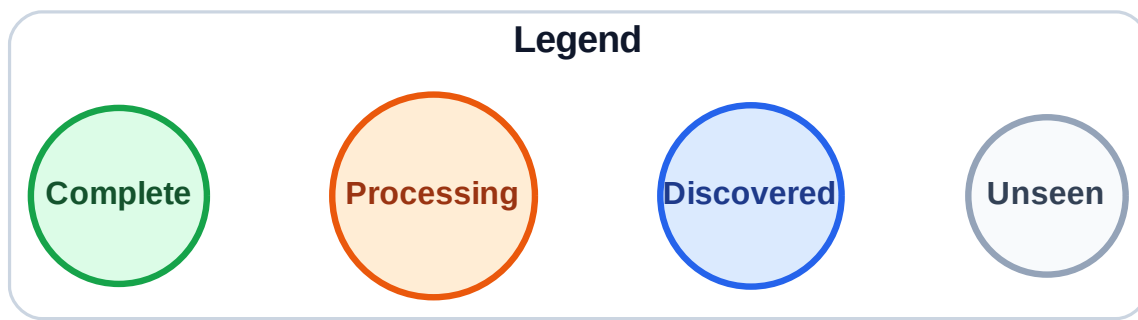


Queue: G → H → B → I → A → J

Visited: C, D, E, F

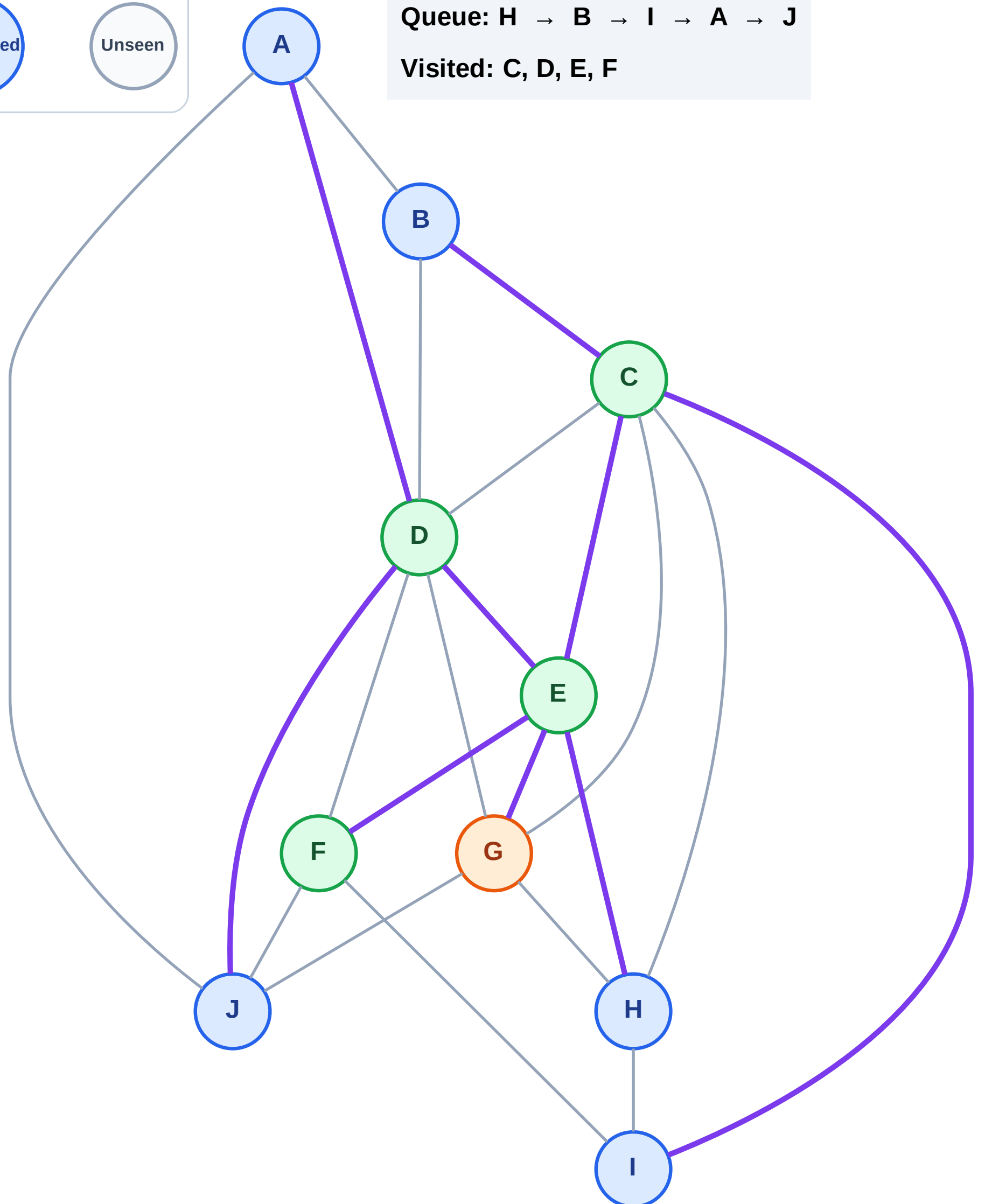


Step 10: Process node G



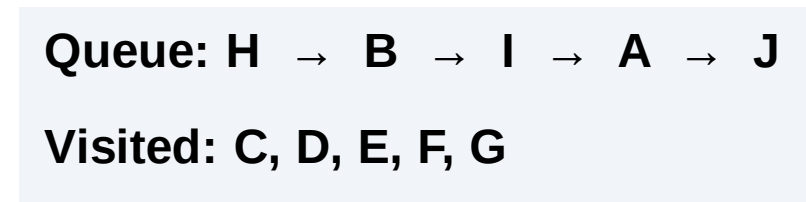
Queue: H → B → I → A → J

Visited: C, D, E, F



Step 11: No new neighbors from G

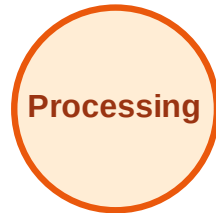
Step 11: No new neighbors from G



BFS Traversal

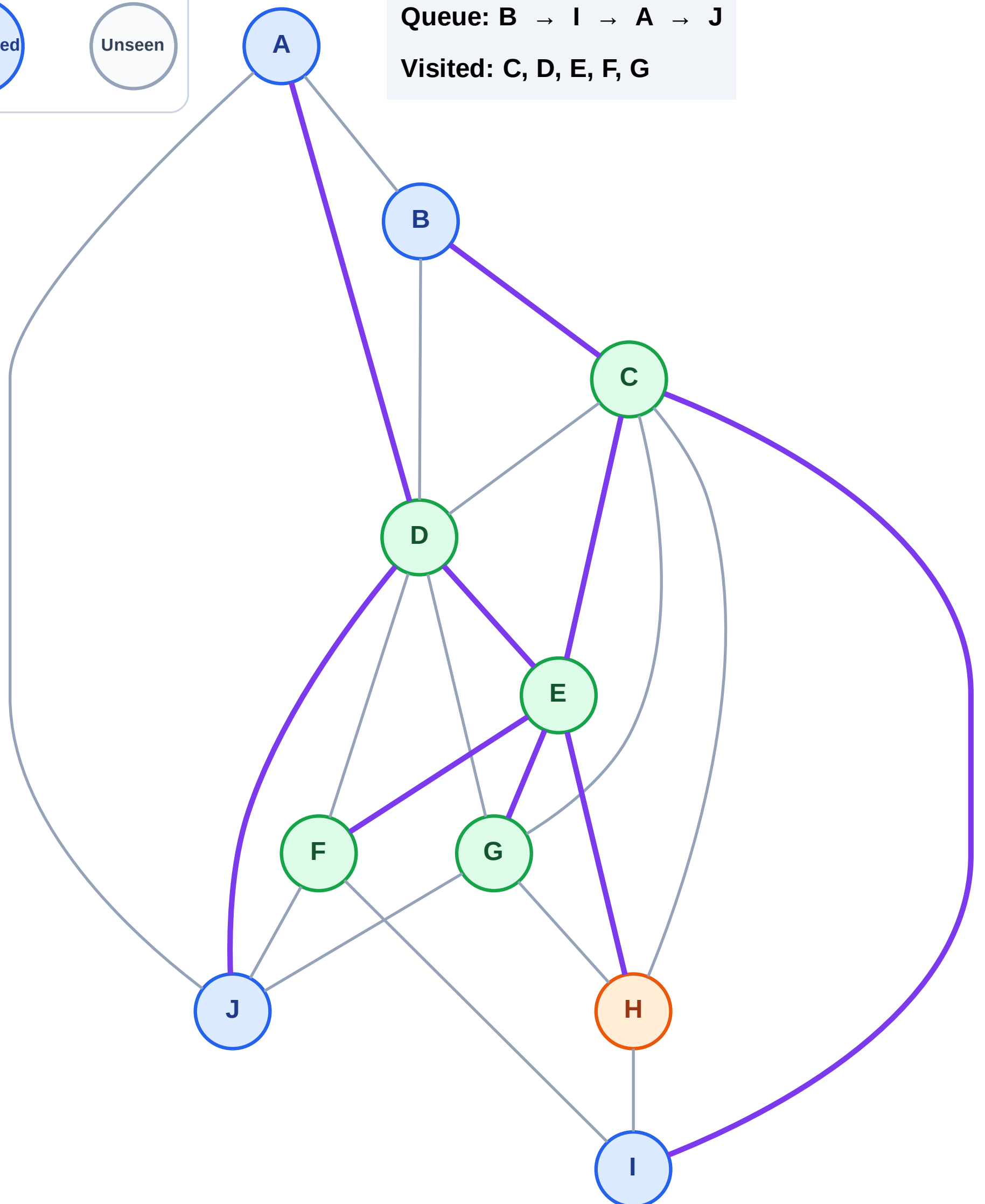
Step 12: Process node H

Legend



Queue: B → I → A → J

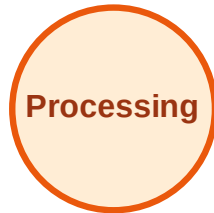
Visited: C, D, E, F, G



BFS Traversal

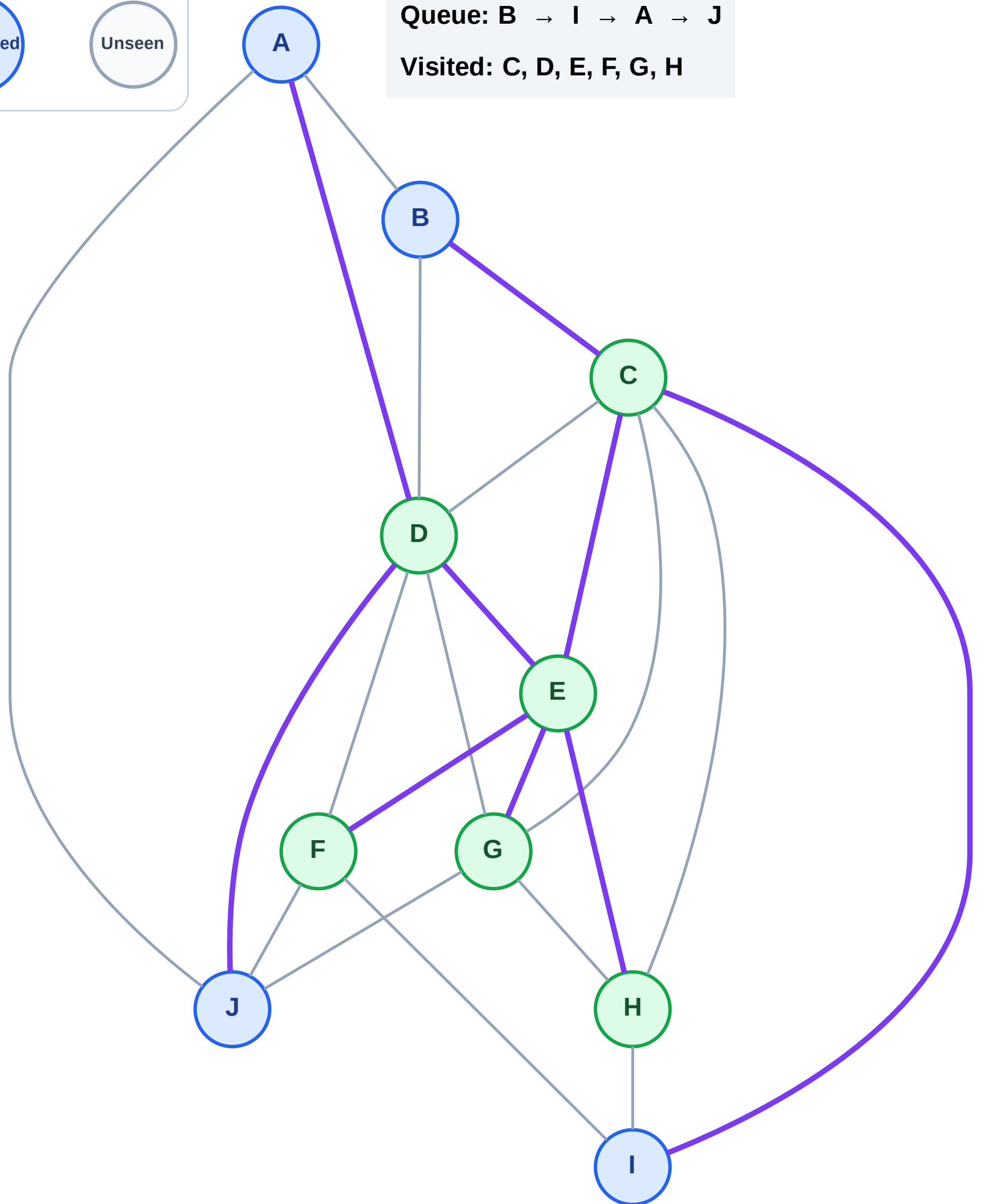
Step 13: No new neighbors from H

Legend



Queue: B → I → A → J

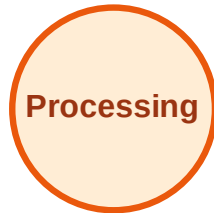
Visited: C, D, E, F, G, H



BFS Traversal

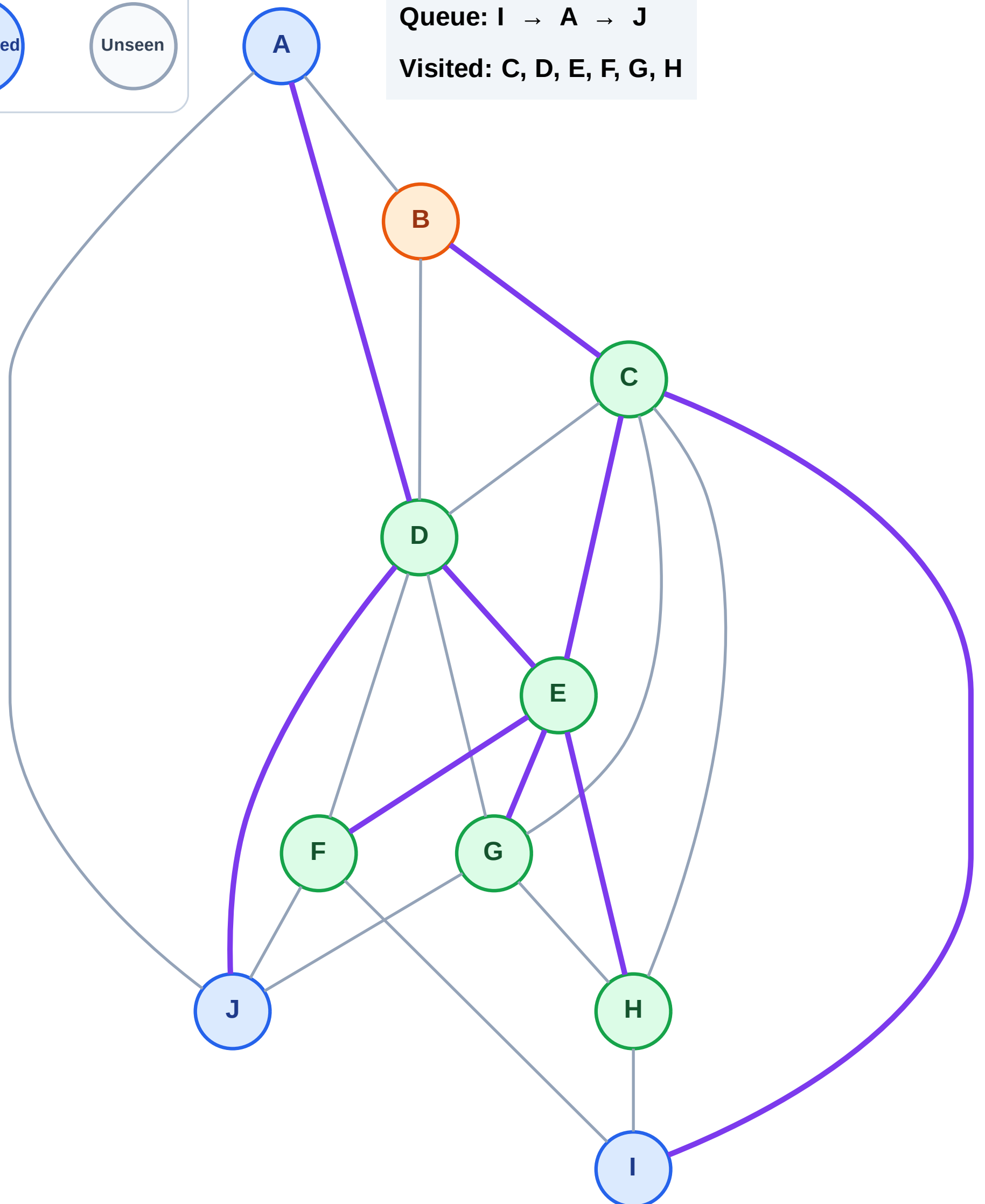
Step 14: Process node B

Legend



Queue: I → A → J

Visited: C, D, E, F, G, H



BFS Traversal

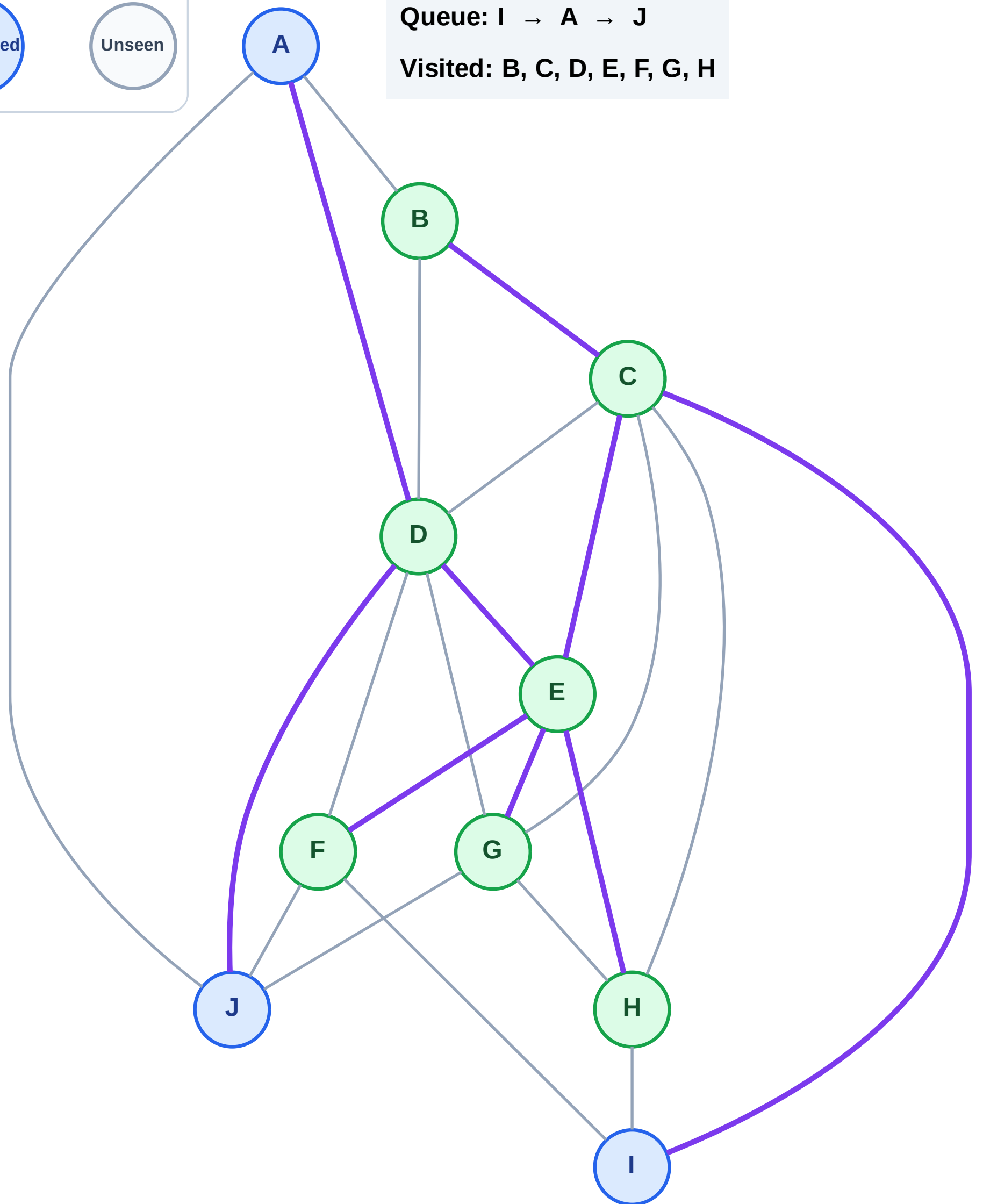
Step 15: No new neighbors from B

Legend



Queue: I → A → J

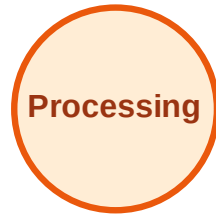
Visited: B, C, D, E, F, G, H



BFS Traversal

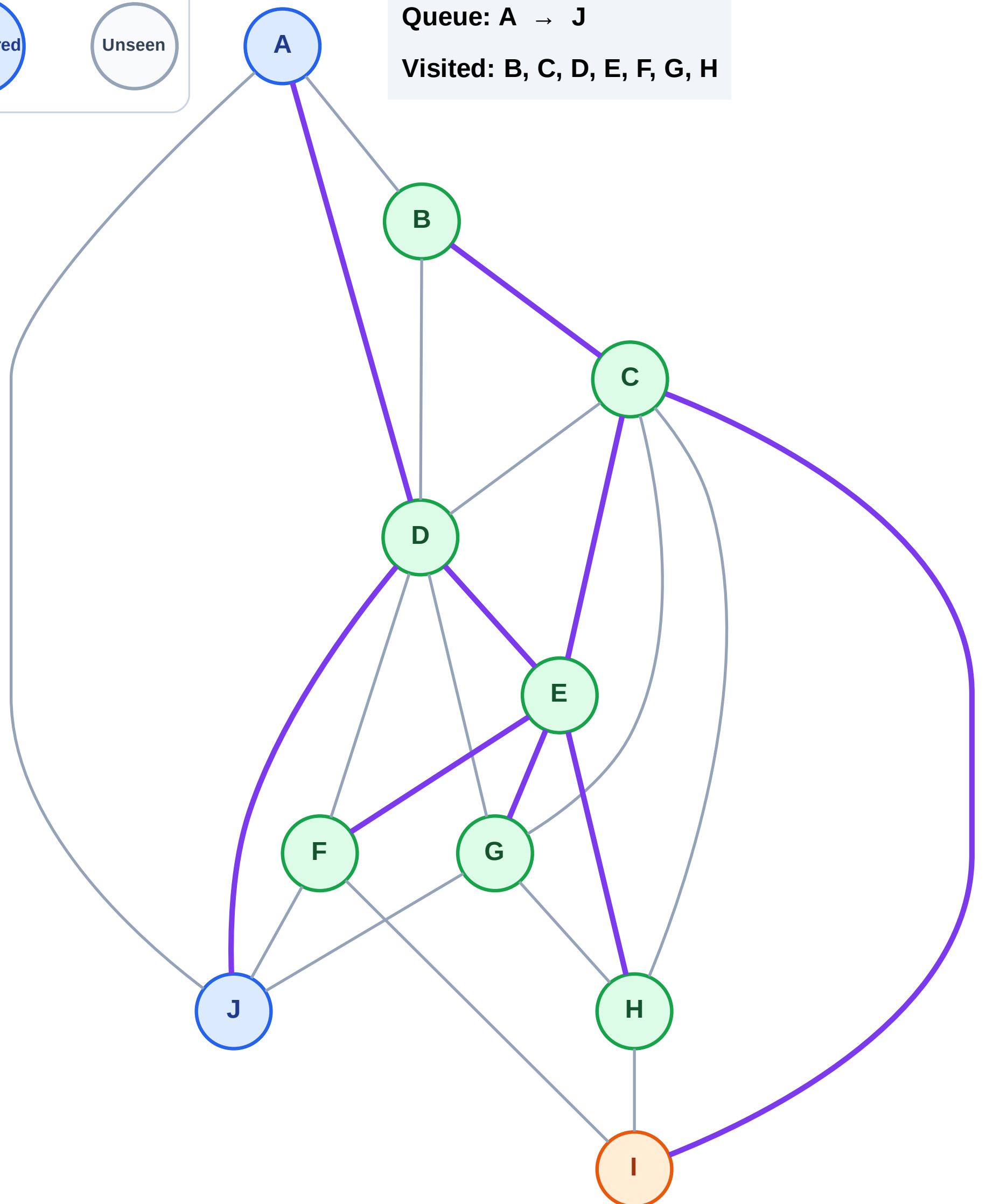
Step 16: Process node I

Legend



Queue: A → J

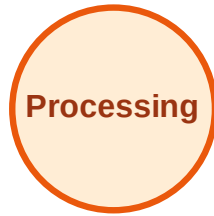
Visited: B, C, D, E, F, G, H



BFS Traversal

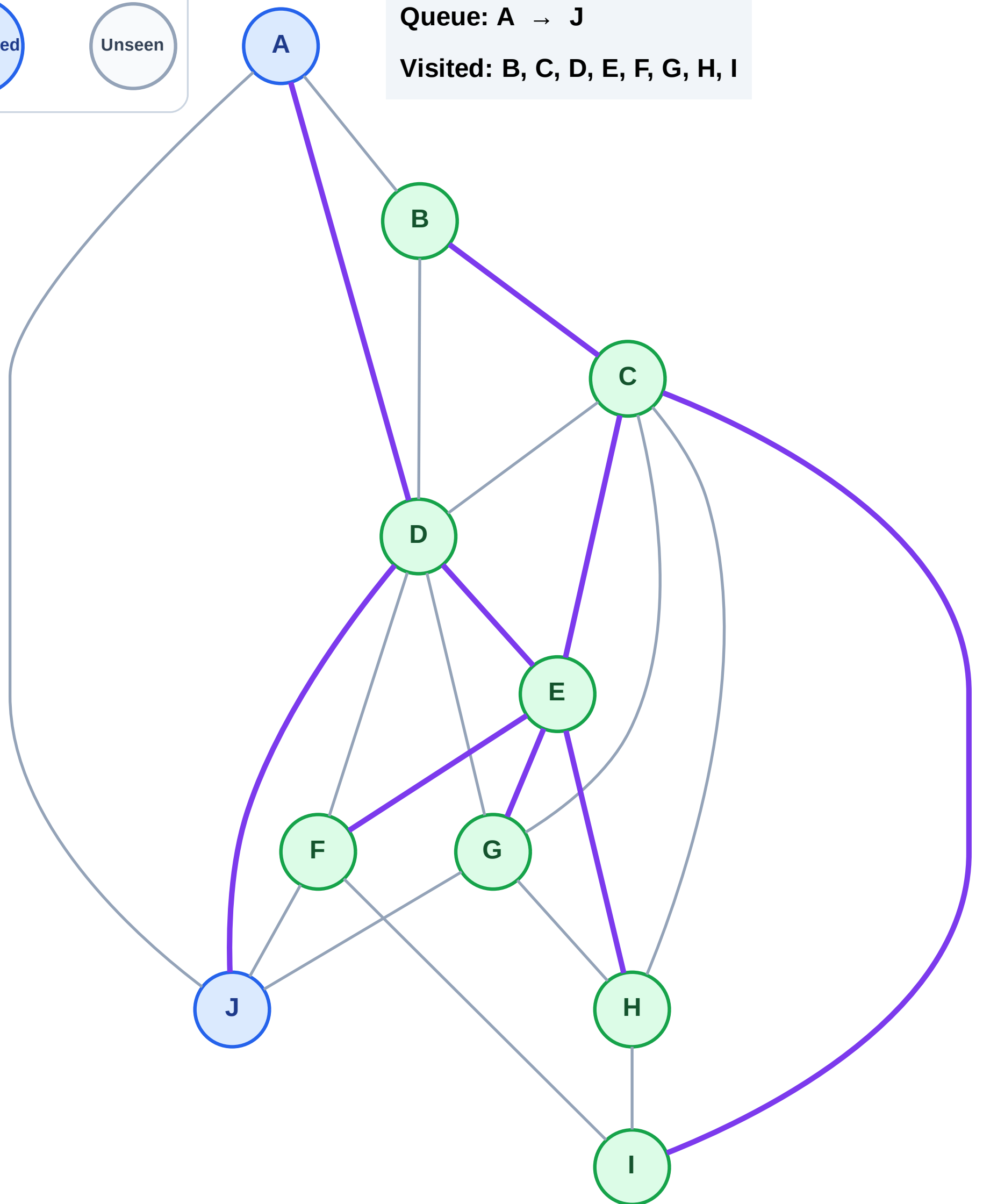
Step 17: No new neighbors from I

Legend



Queue: A → J

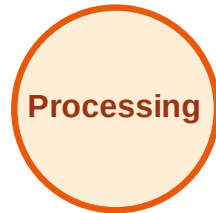
Visited: B, C, D, E, F, G, H, I



BFS Traversal

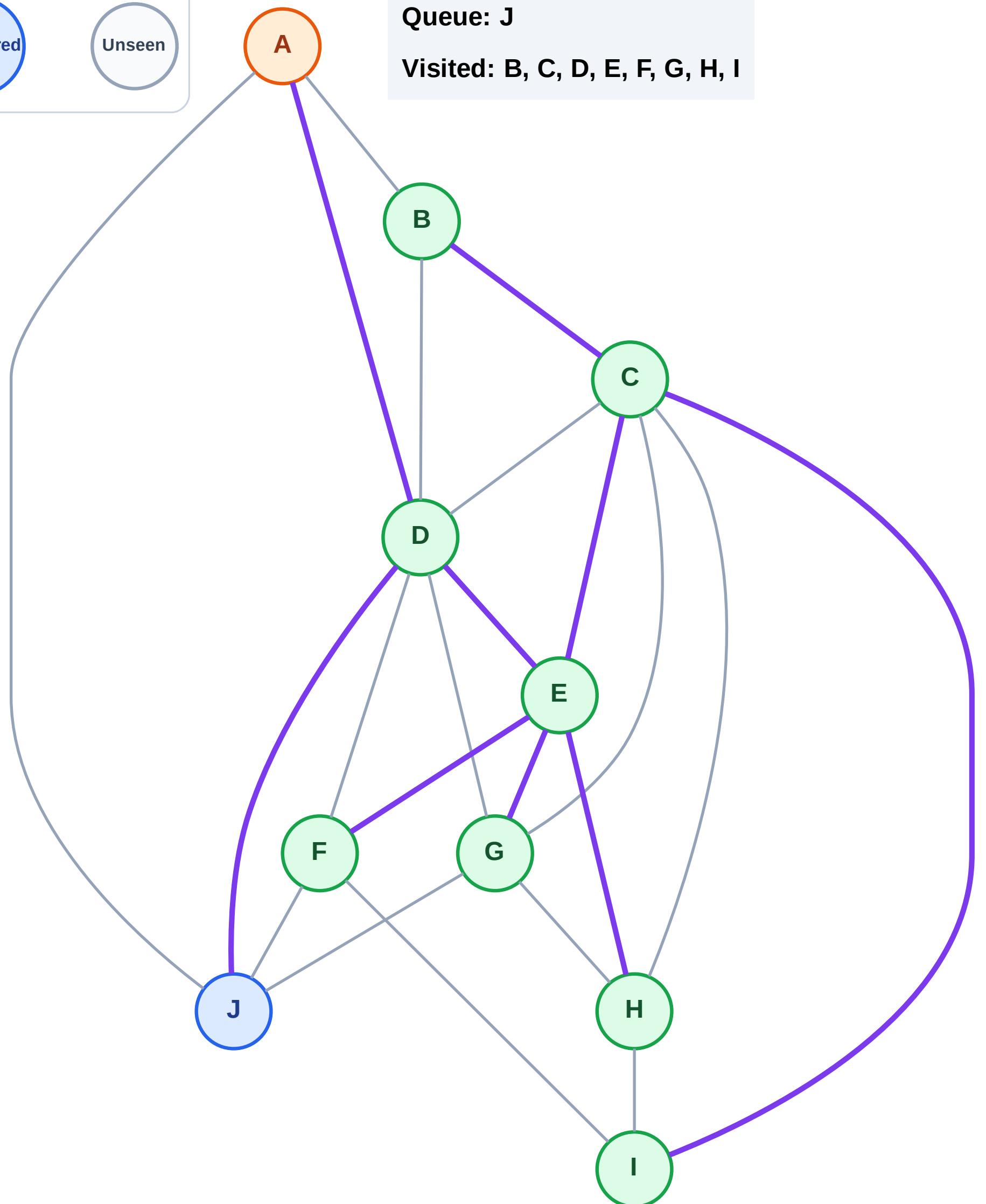
Step 18: Process node A

Legend



Queue: J

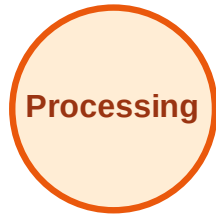
Visited: B, C, D, E, F, G, H, I



BFS Traversal

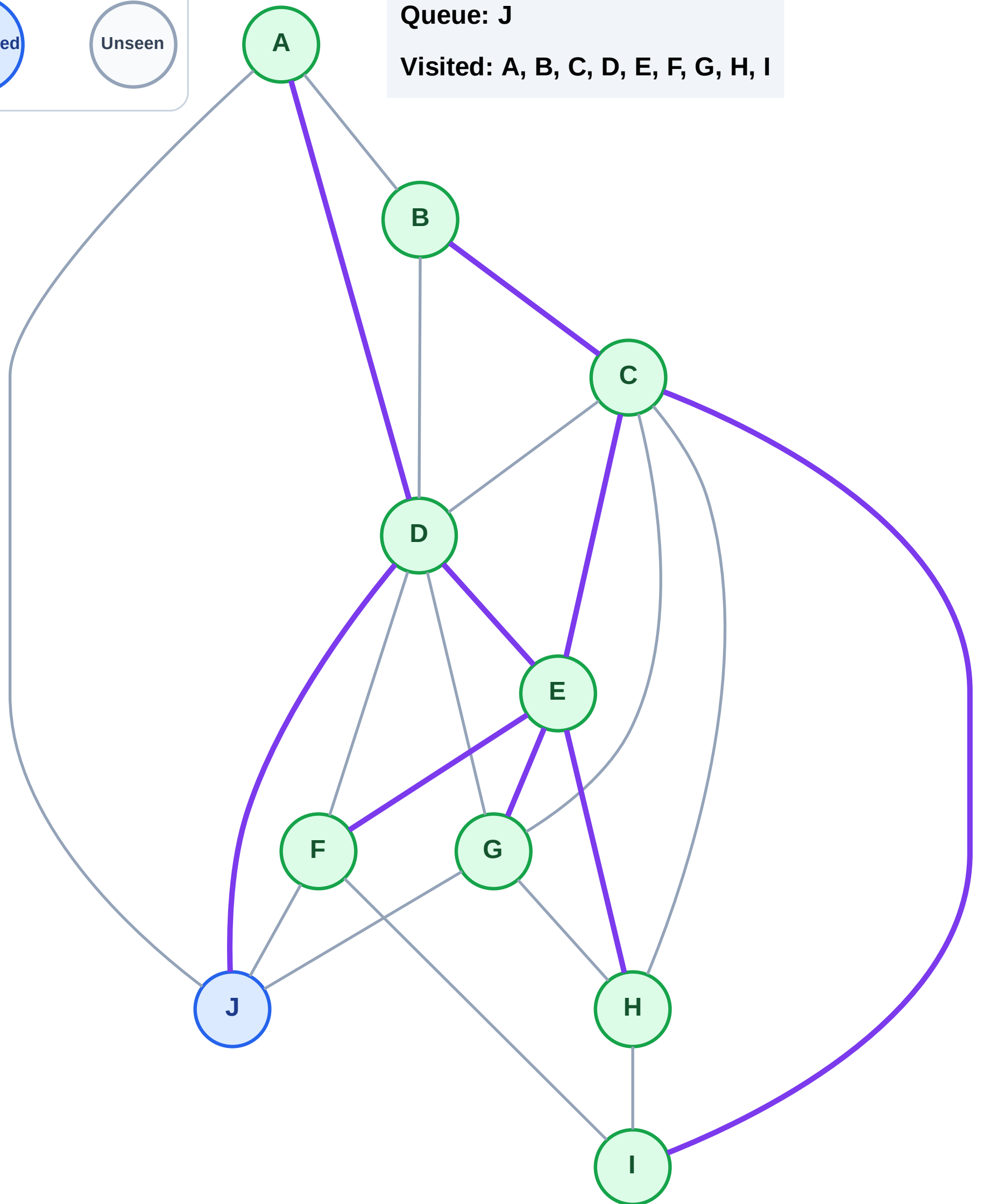
Step 19: No new neighbors from A

Legend



Queue: J

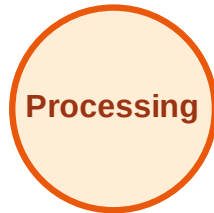
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

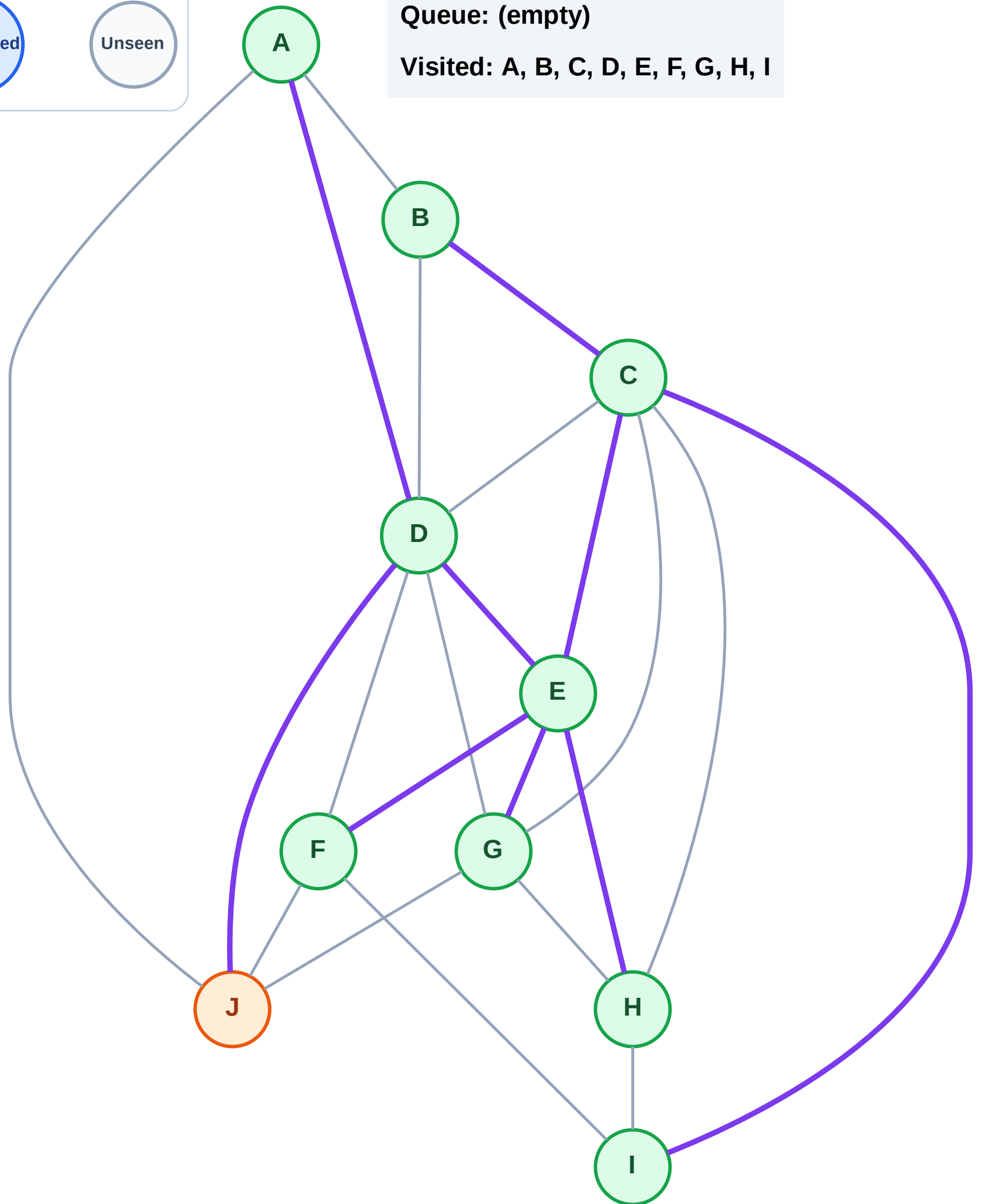
Step 20: Process node J

Legend



Queue: (empty)

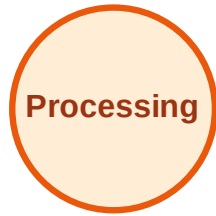
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

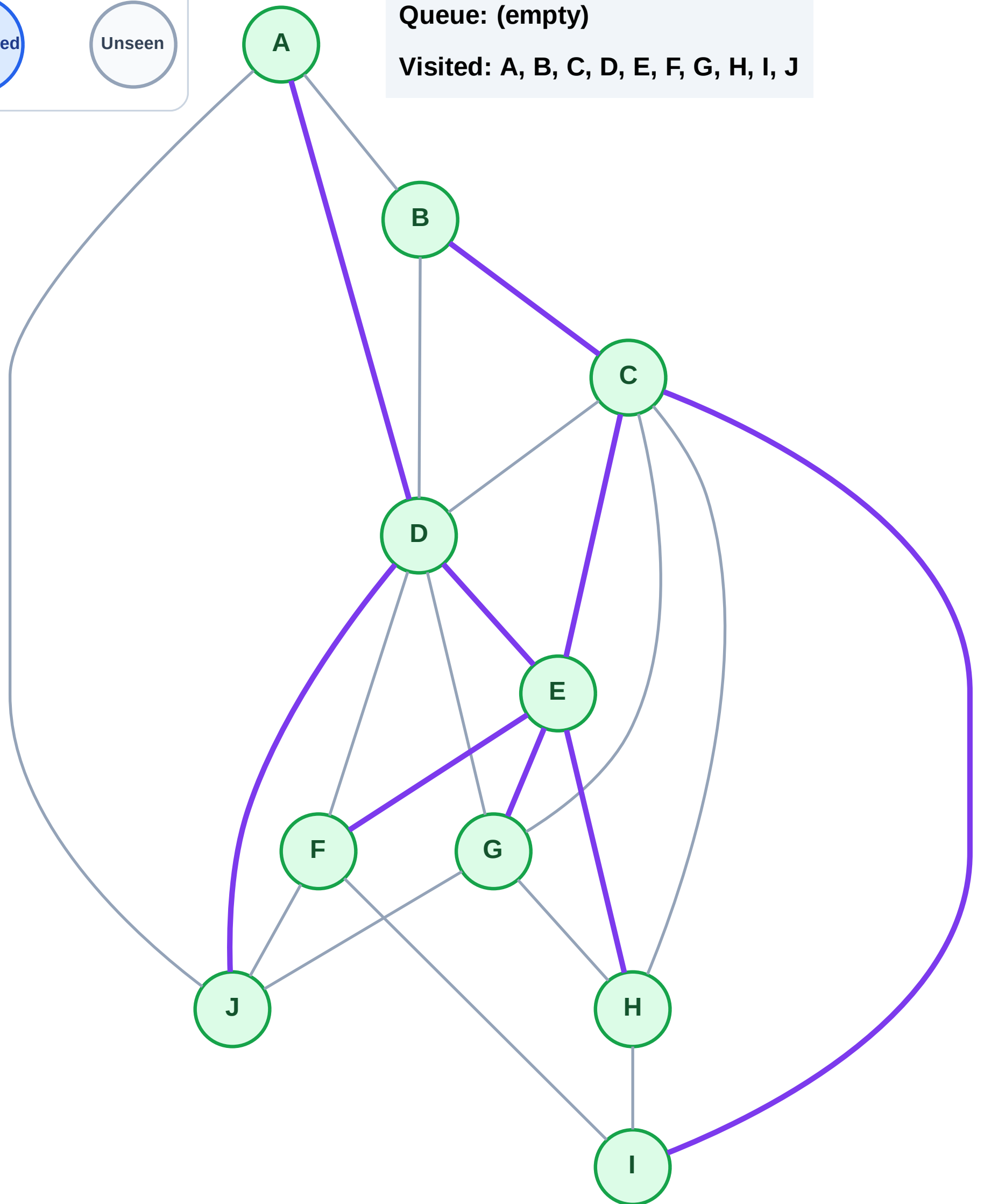
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

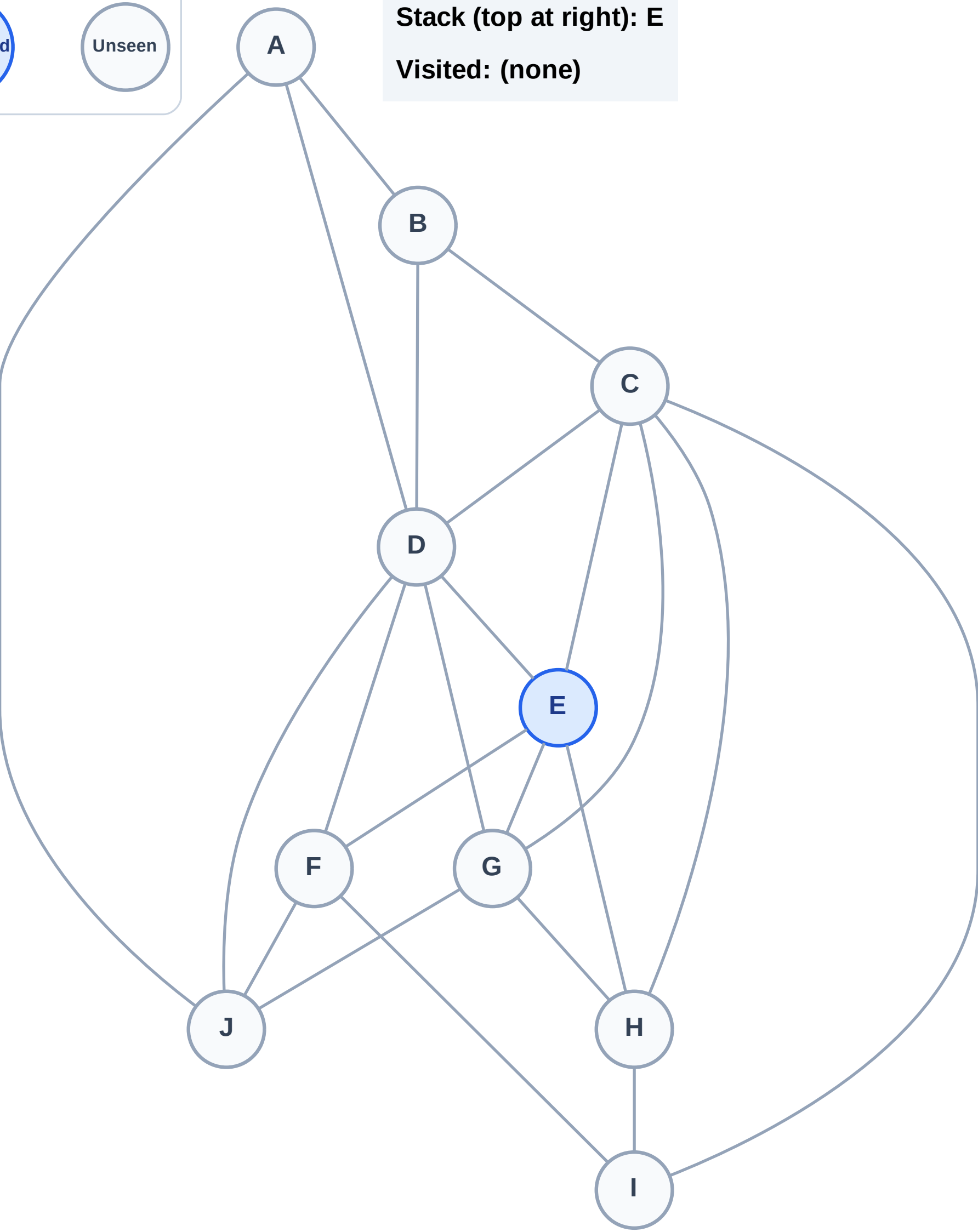
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

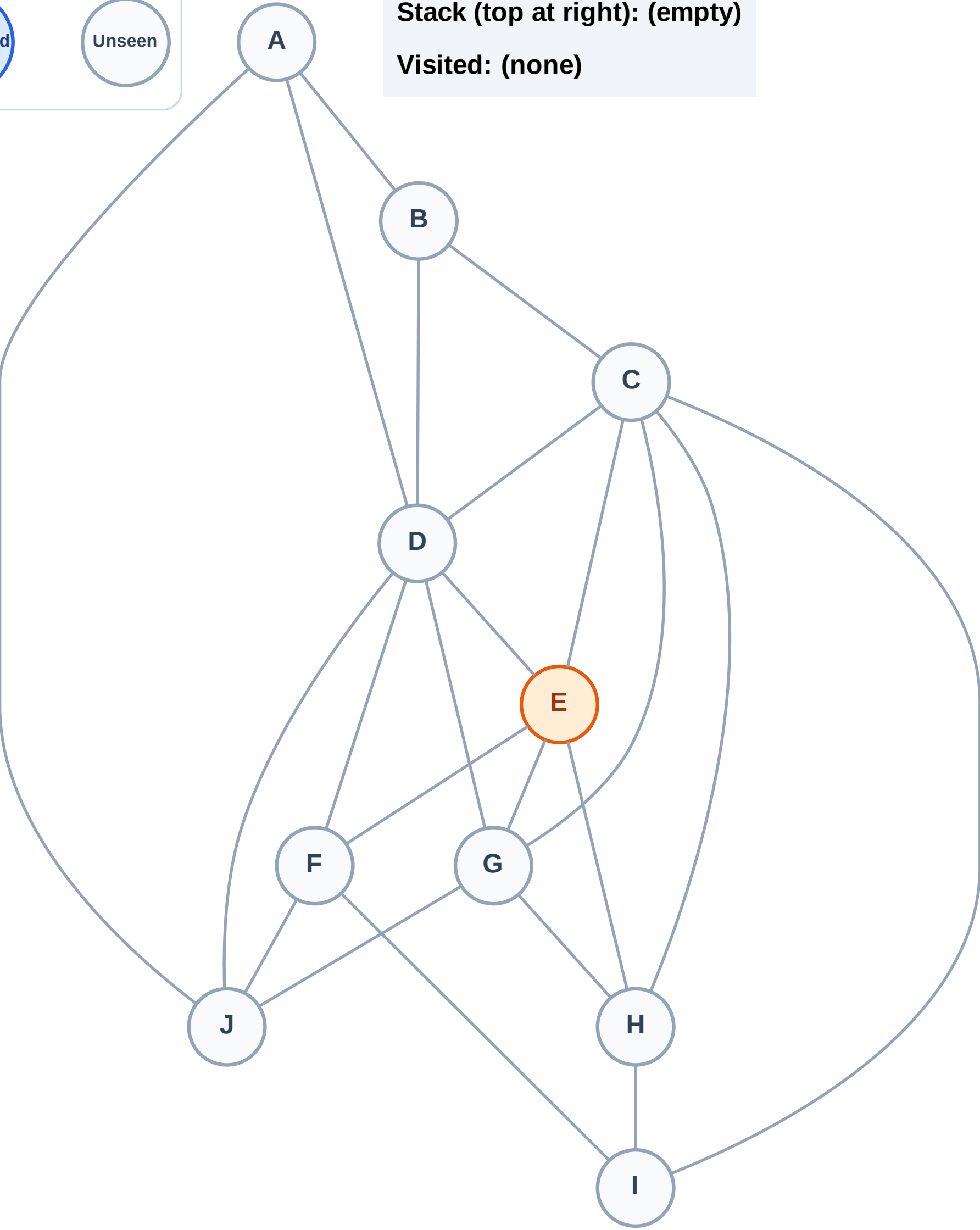
Complete

Processing

Discovered

Unseen

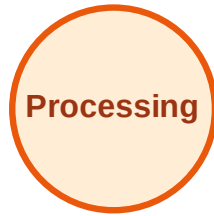
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

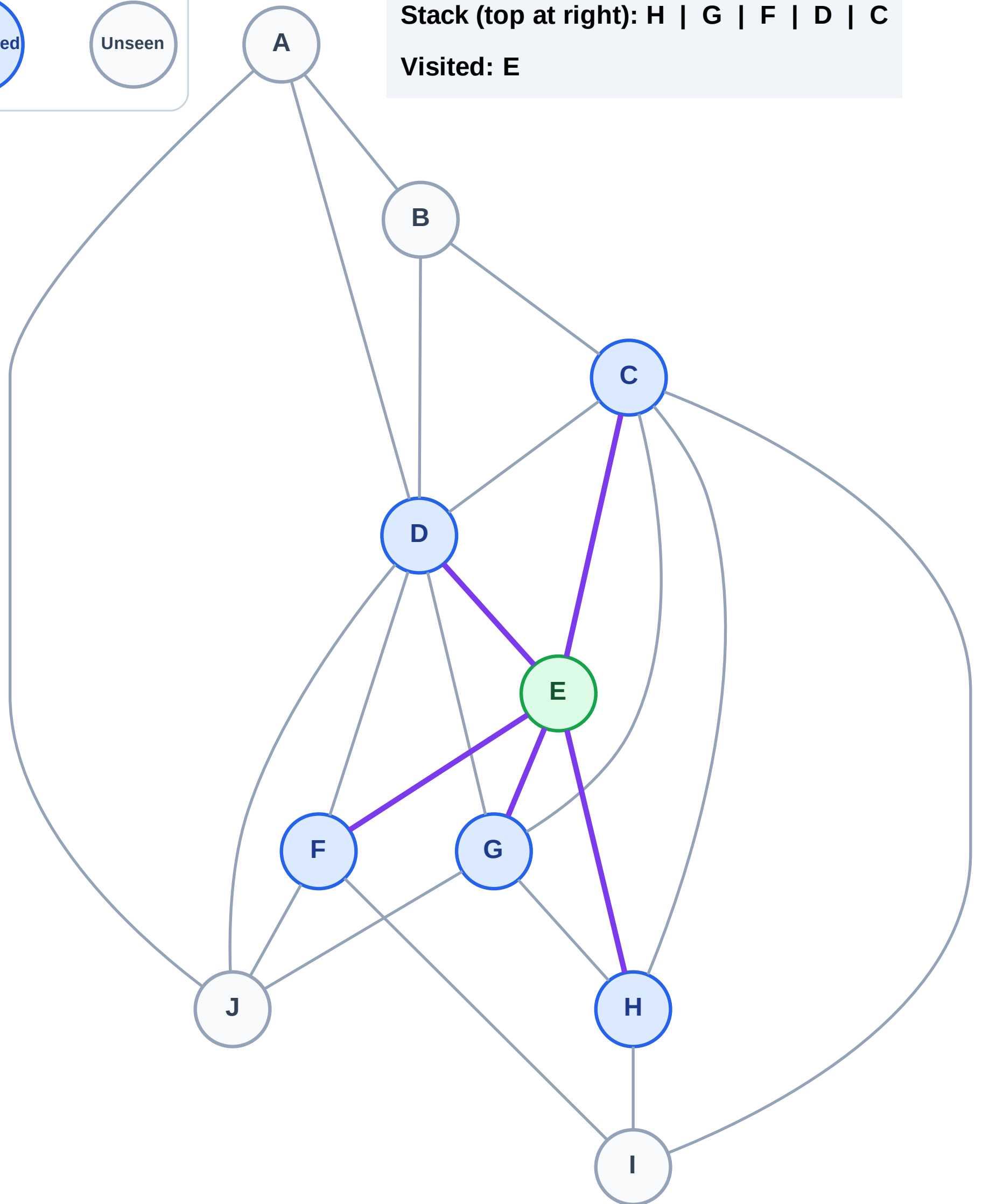
Step 3: Discover H, G, F, D, C from E

Legend



Stack (top at right): H | G | F | D | C

Visited: E



DFS Traversal

Step 4: Process node C

Legend

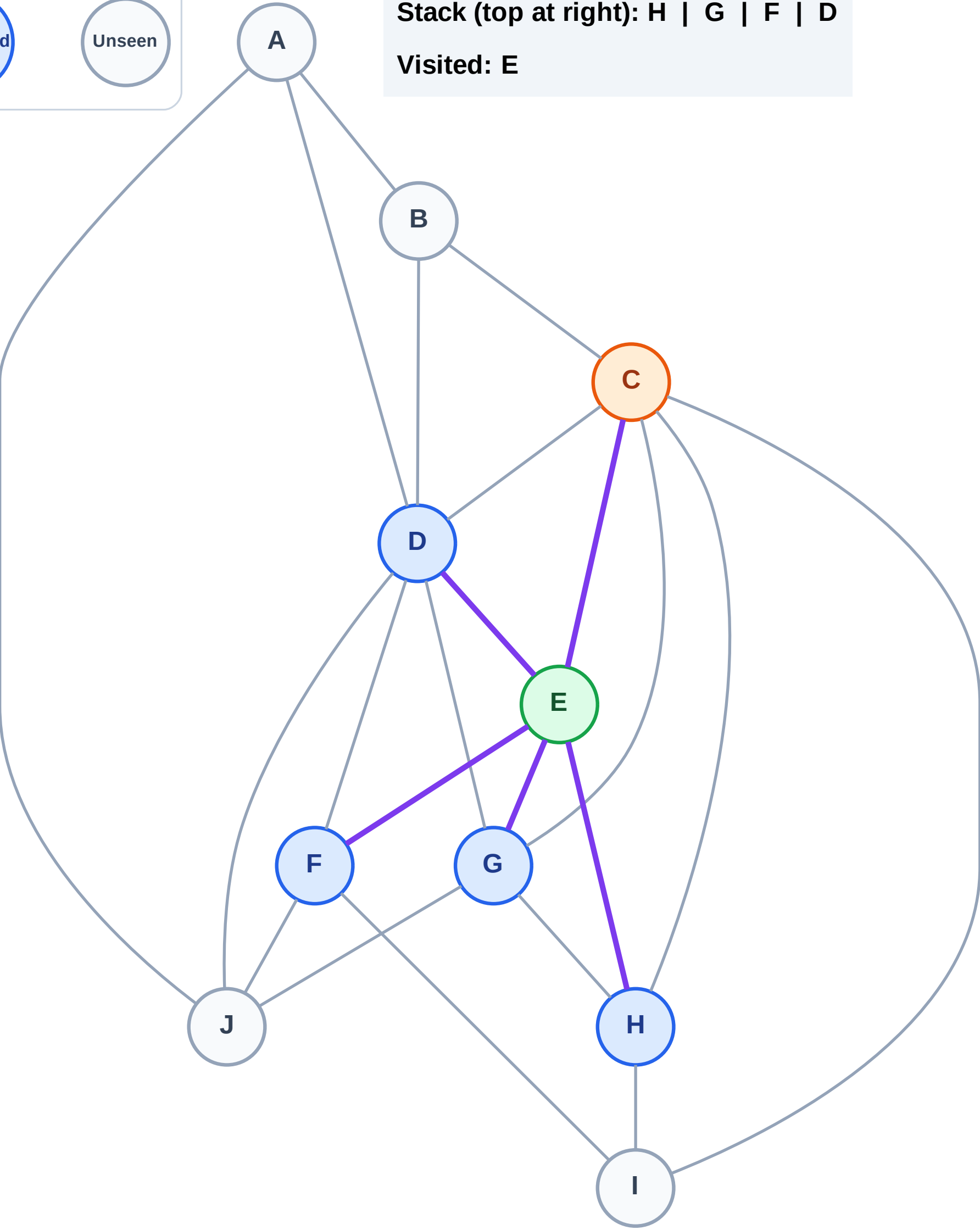
Complete

Processing

Discovered

Unseen

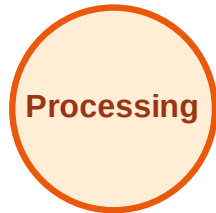
Stack (top at right): H | G | F | D
Visited: E



DFS Traversal

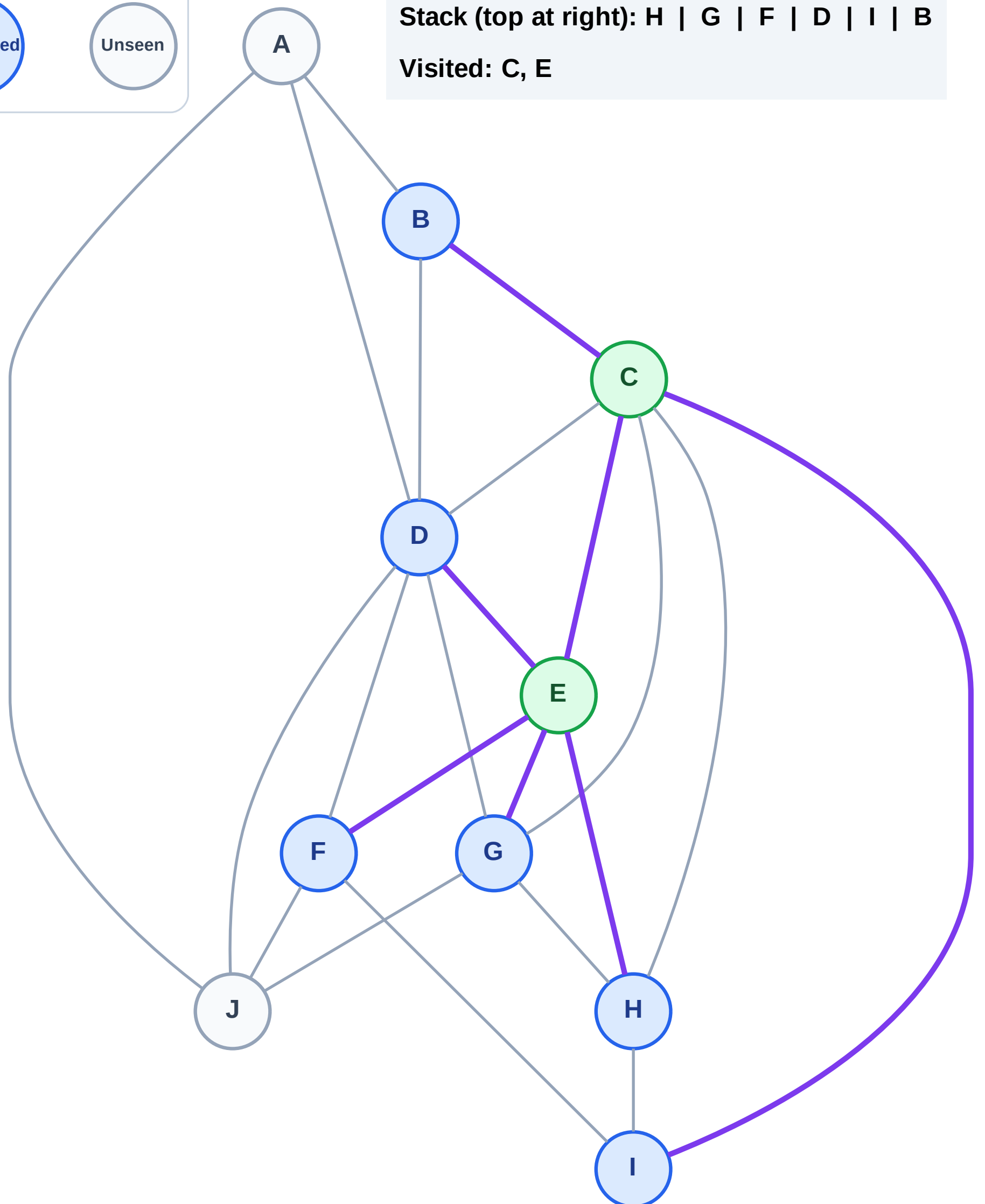
Step 5: Discover I, B from C

Legend



Stack (top at right): H | G | F | D | I | B

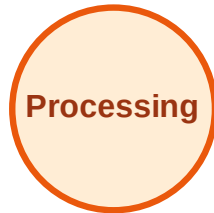
Visited: C, E



DFS Traversal

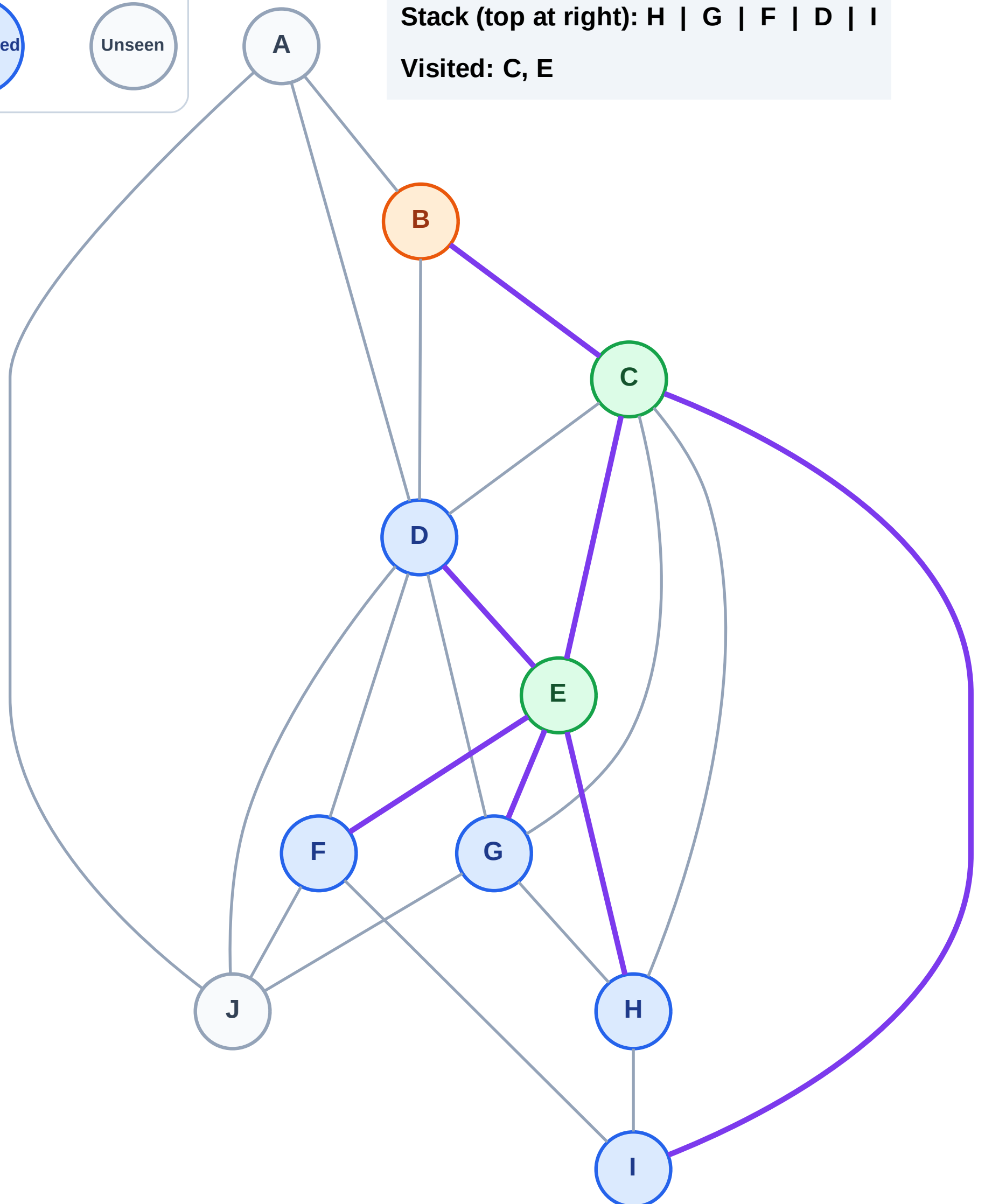
Step 6: Process node B

Legend



Stack (top at right): H | G | F | D | I

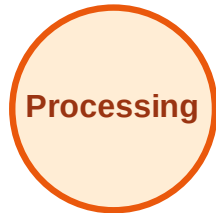
Visited: C, E



DFS Traversal

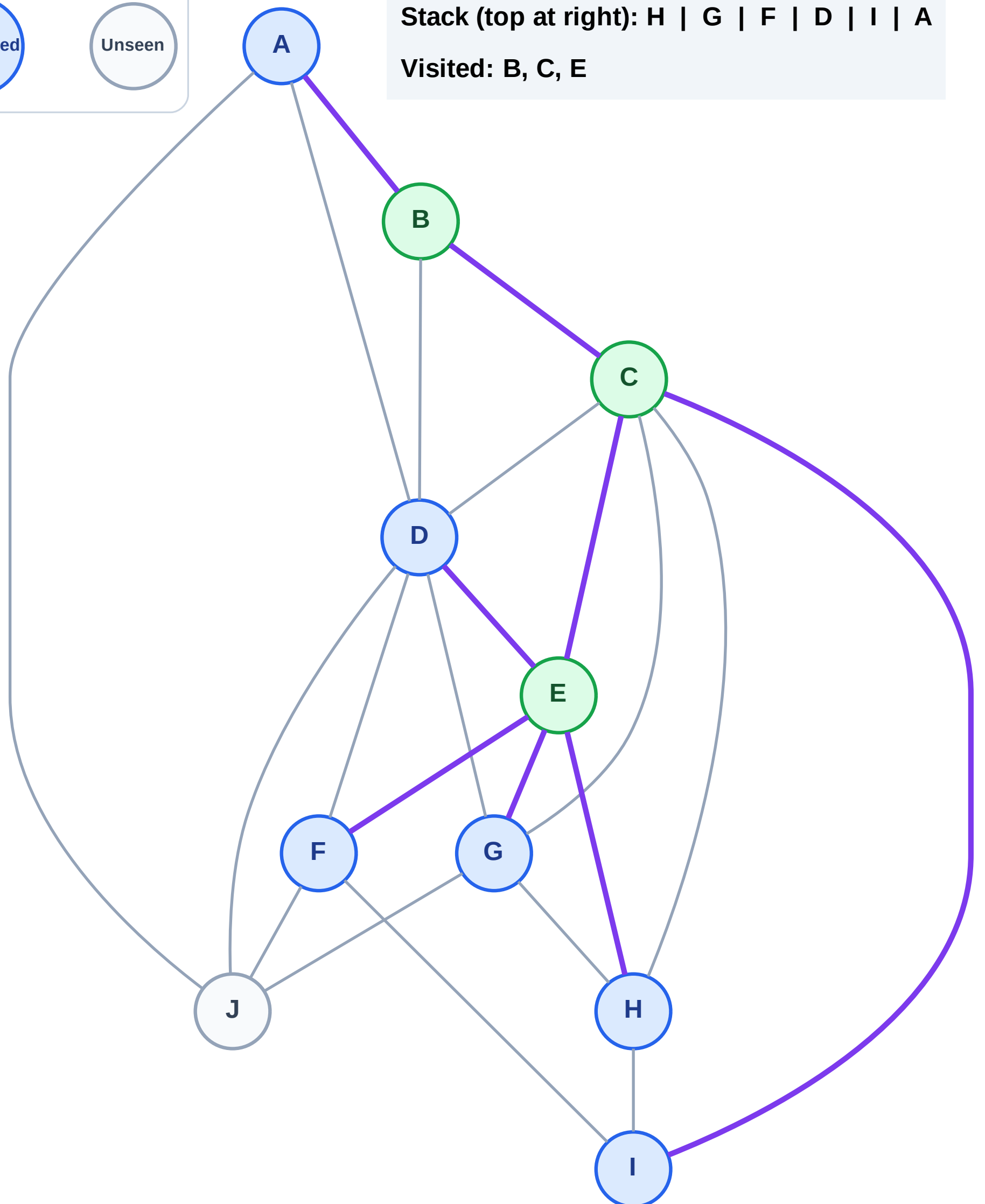
Step 7: Discover A from B

Legend



Stack (top at right): H | G | F | D | I | A

Visited: B, C, E



DFS Traversal

Step 8: Process node A

Legend

Complete

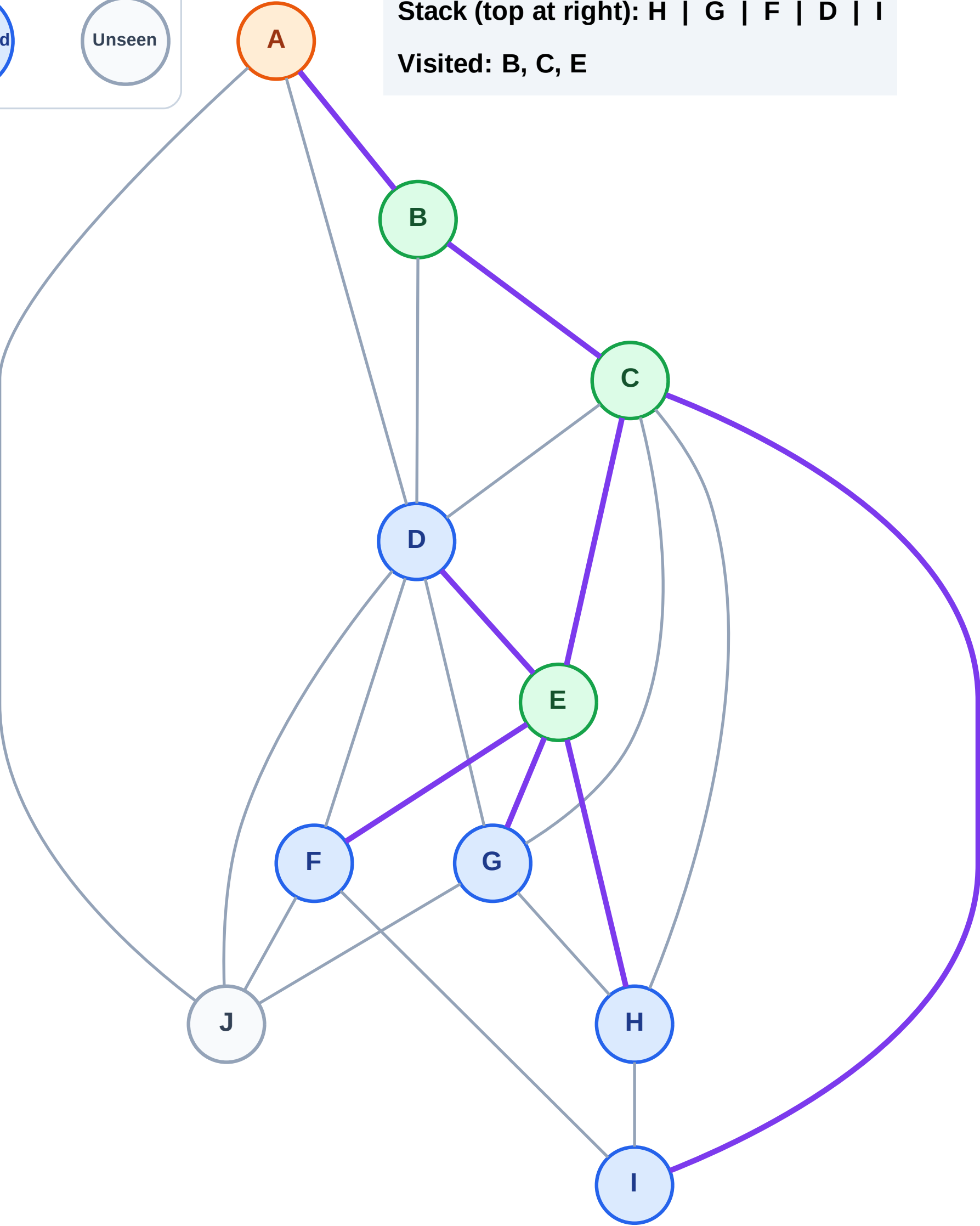
Processing

Discovered

Unseen

Stack (top at right): H | G | F | D | I

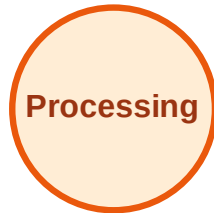
Visited: B, C, E



DFS Traversal

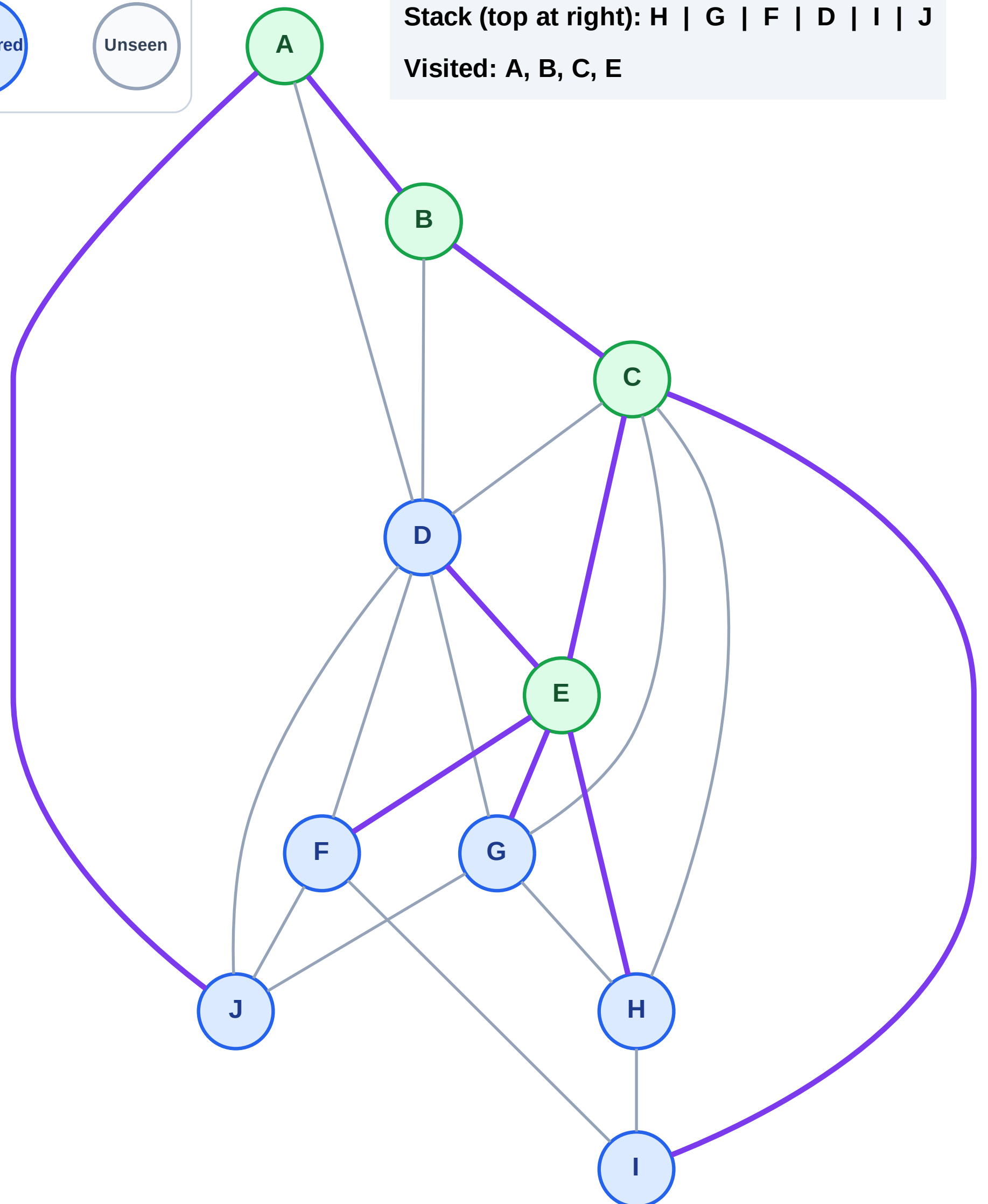
Step 9: Discover J from A

Legend



Stack (top at right): H | G | F | D | I | J

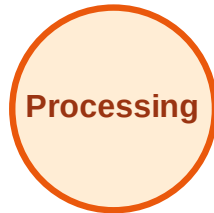
Visited: A, B, C, E



DFS Traversal

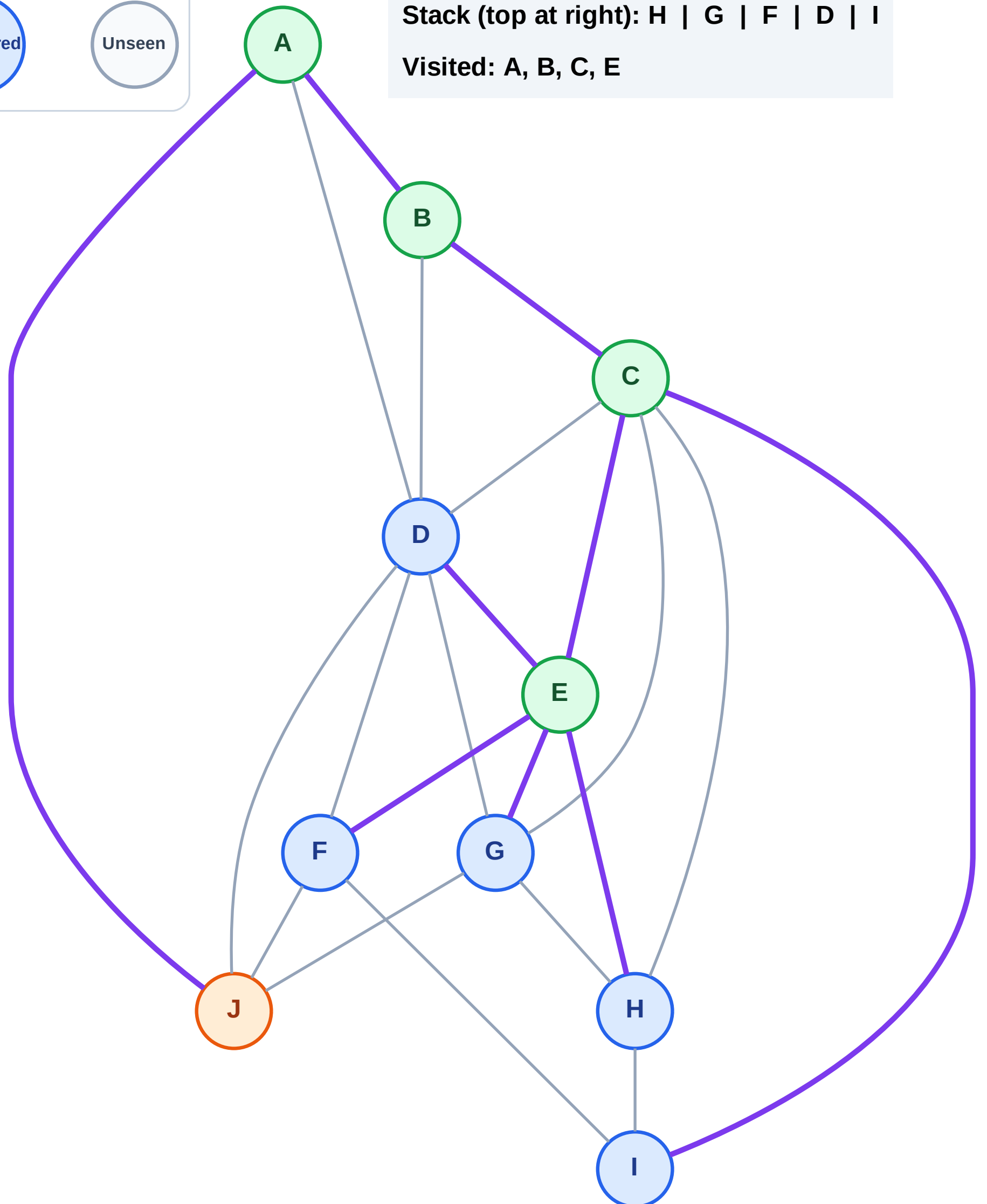
Step 10: Process node J

Legend



Stack (top at right): H | G | F | D | I

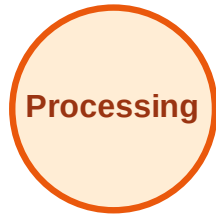
Visited: A, B, C, E



DFS Traversal

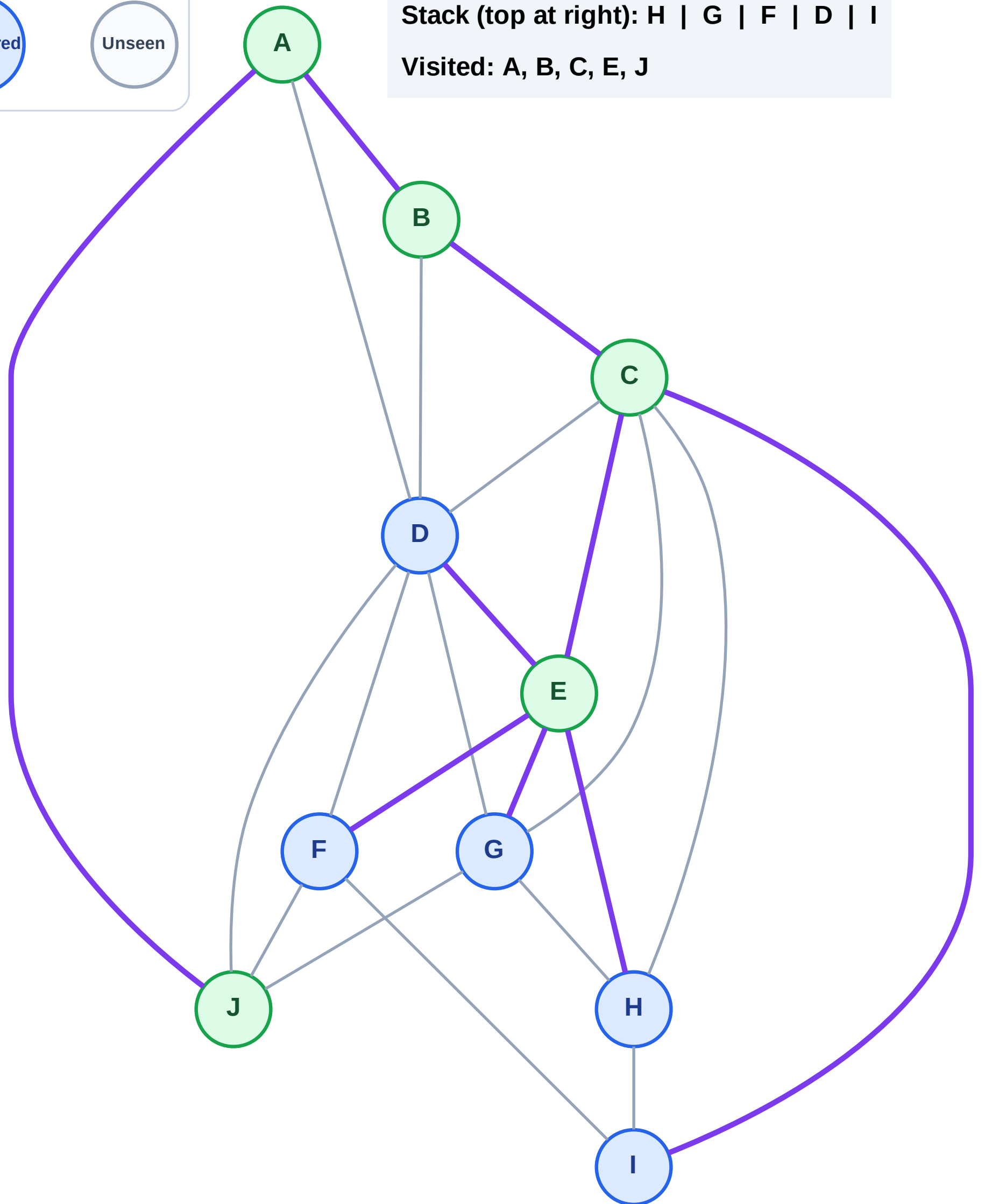
Step 11: No new neighbors from J

Legend



Stack (top at right): H | G | F | D | I

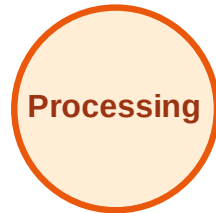
Visited: A, B, C, E, J



DFS Traversal

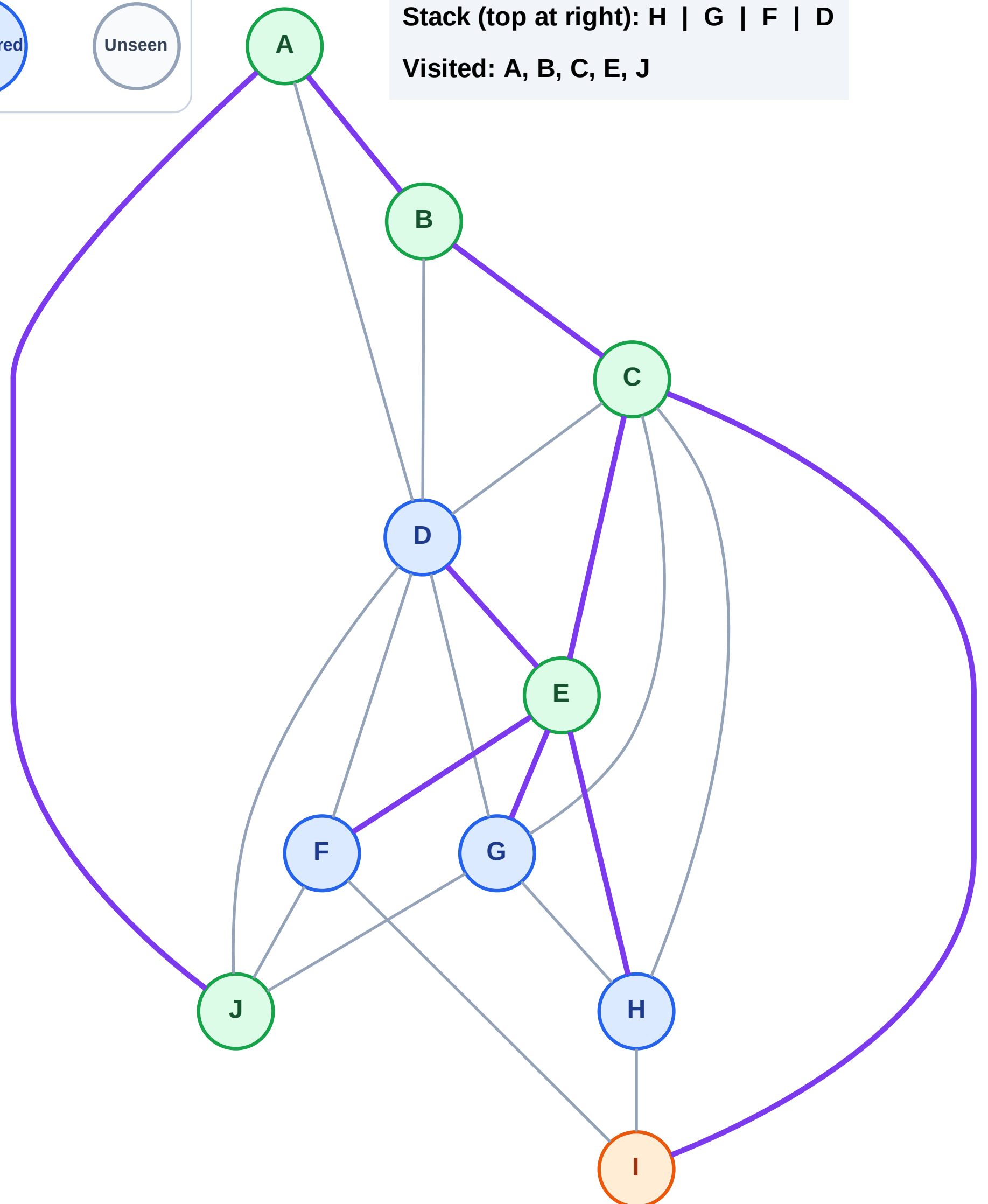
Step 12: Process node I

Legend



Stack (top at right): H | G | F | D

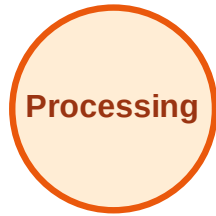
Visited: A, B, C, E, J



DFS Traversal

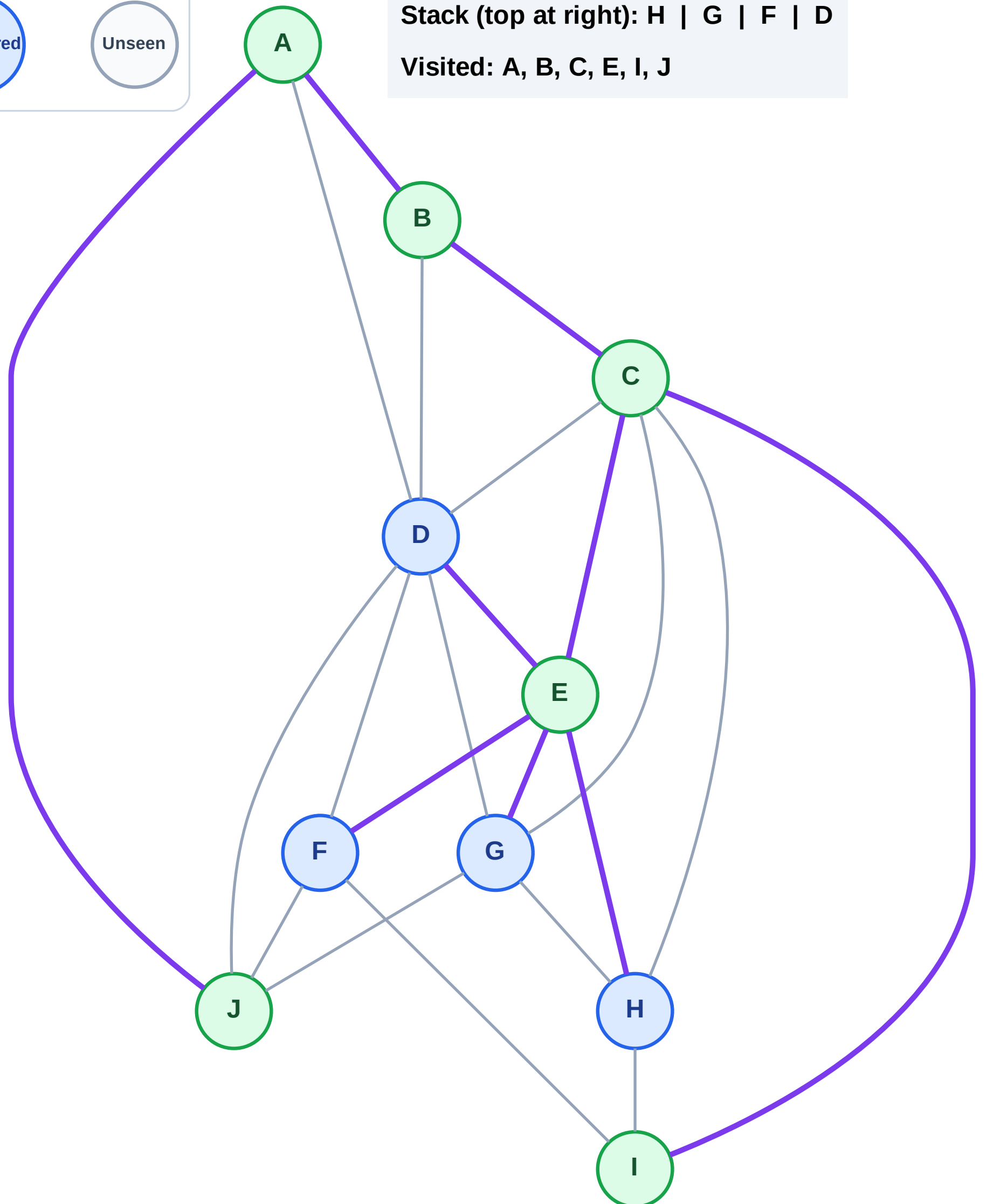
Step 13: No new neighbors from I

Legend



Stack (top at right): H | G | F | D

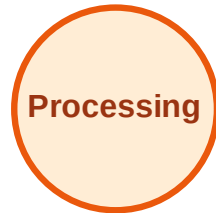
Visited: A, B, C, E, I, J



DFS Traversal

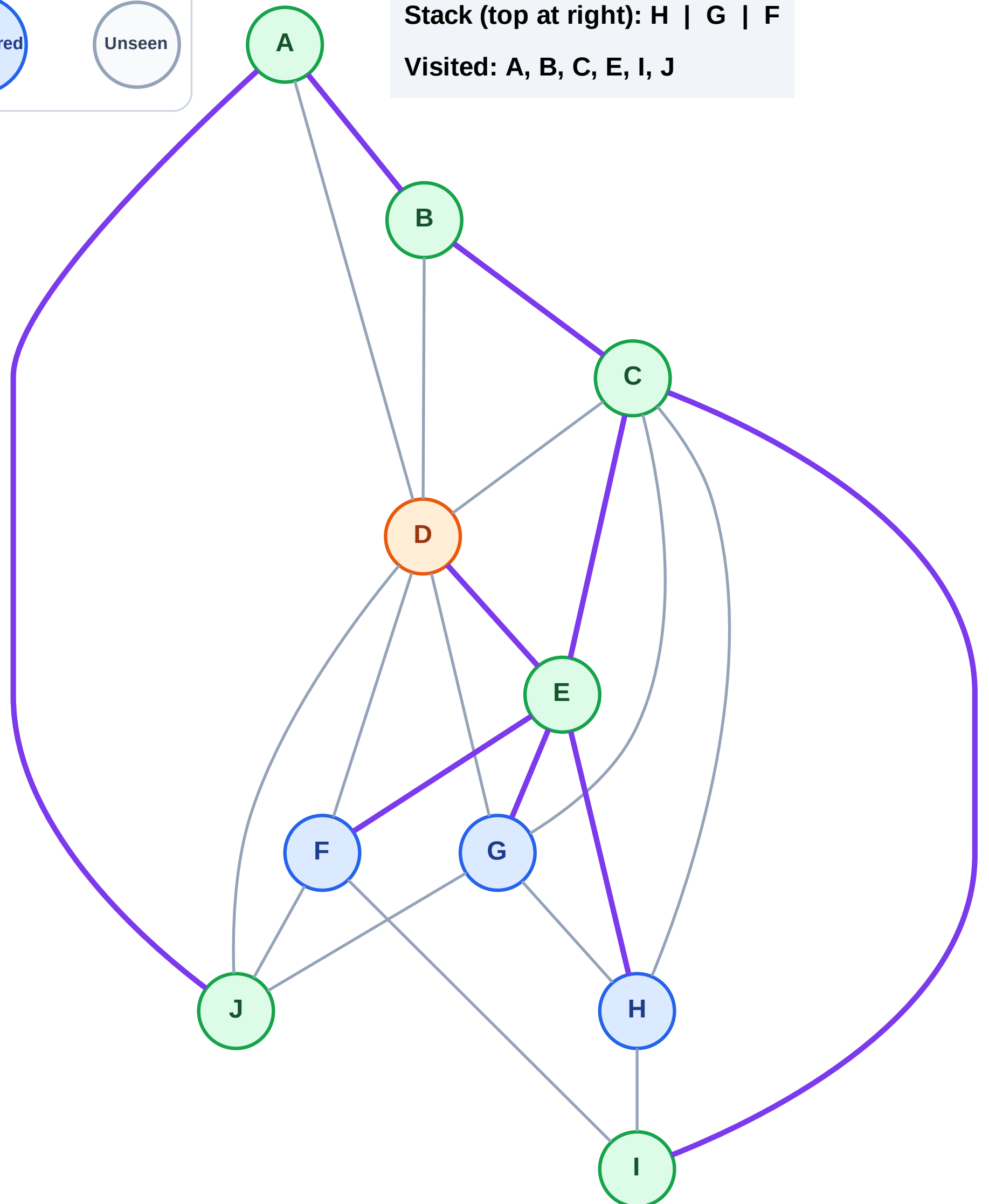
Step 14: Process node D

Legend



Stack (top at right): H | G | F

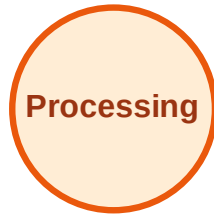
Visited: A, B, C, E, I, J



DFS Traversal

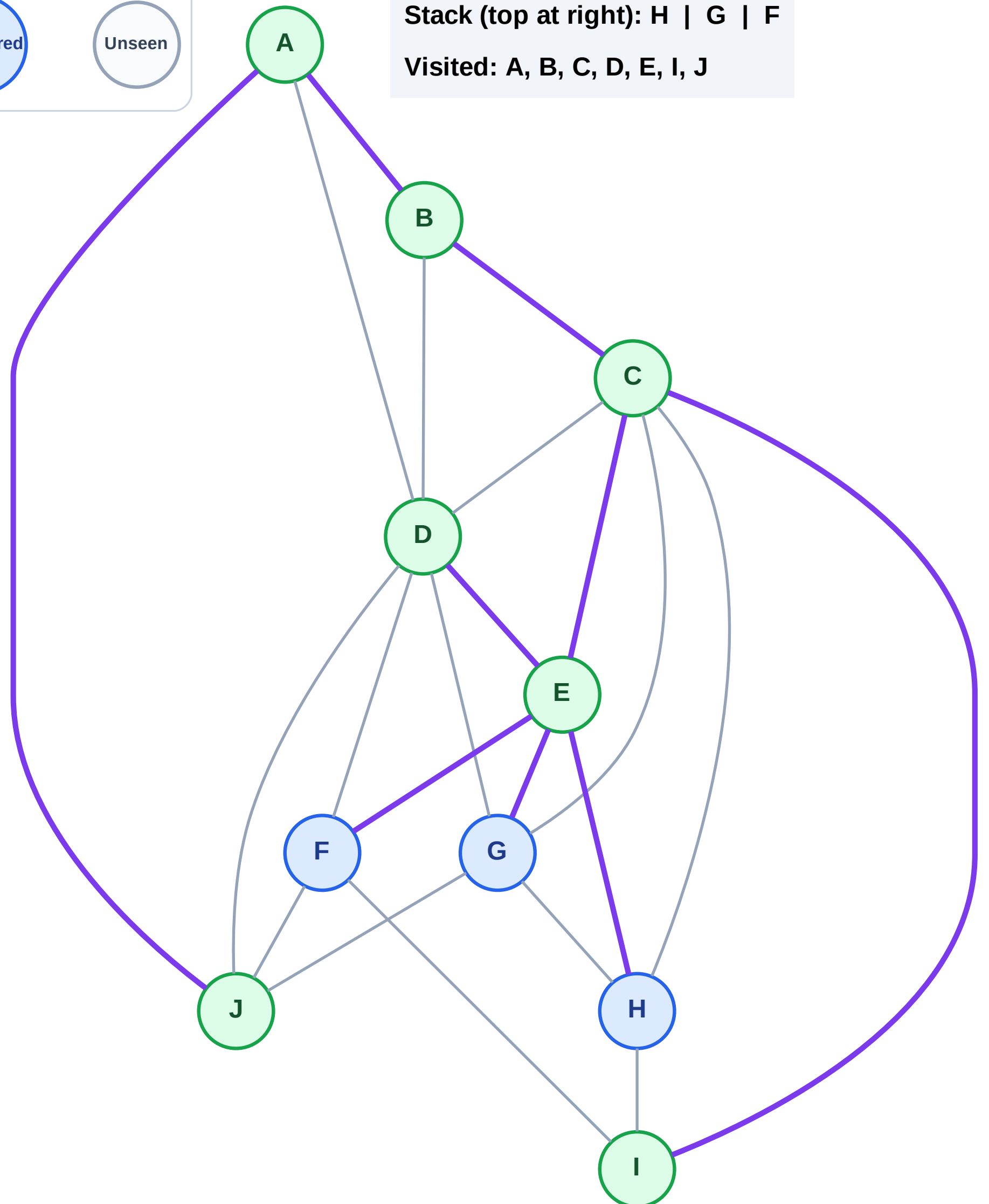
Step 15: No new neighbors from D

Legend



Stack (top at right): H | G | F

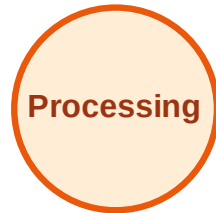
Visited: A, B, C, D, E, I, J



DFS Traversal

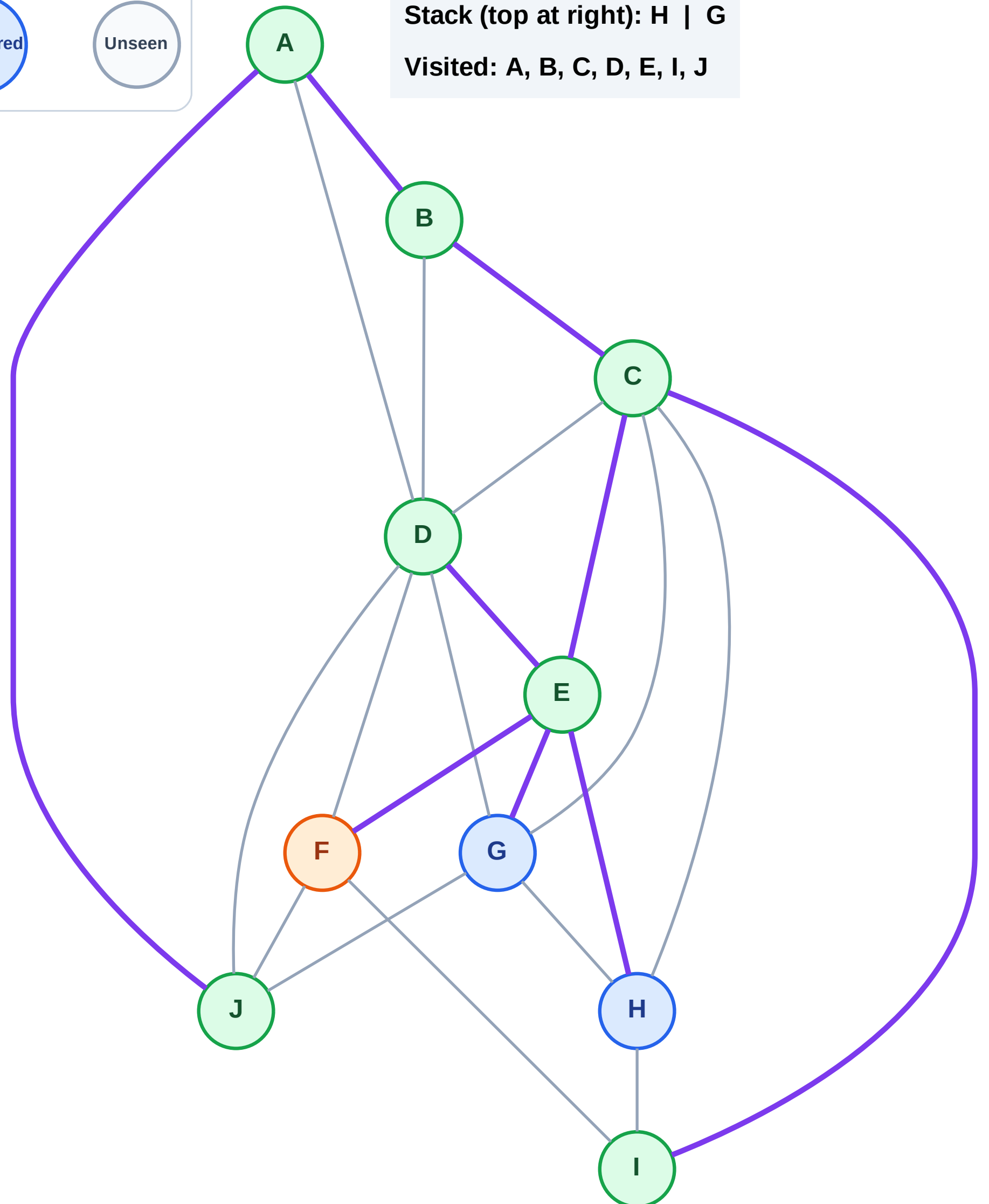
Step 16: Process node F

Legend



Stack (top at right): H | G

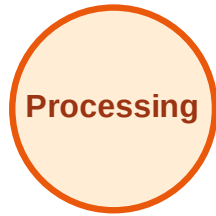
Visited: A, B, C, D, E, I, J



DFS Traversal

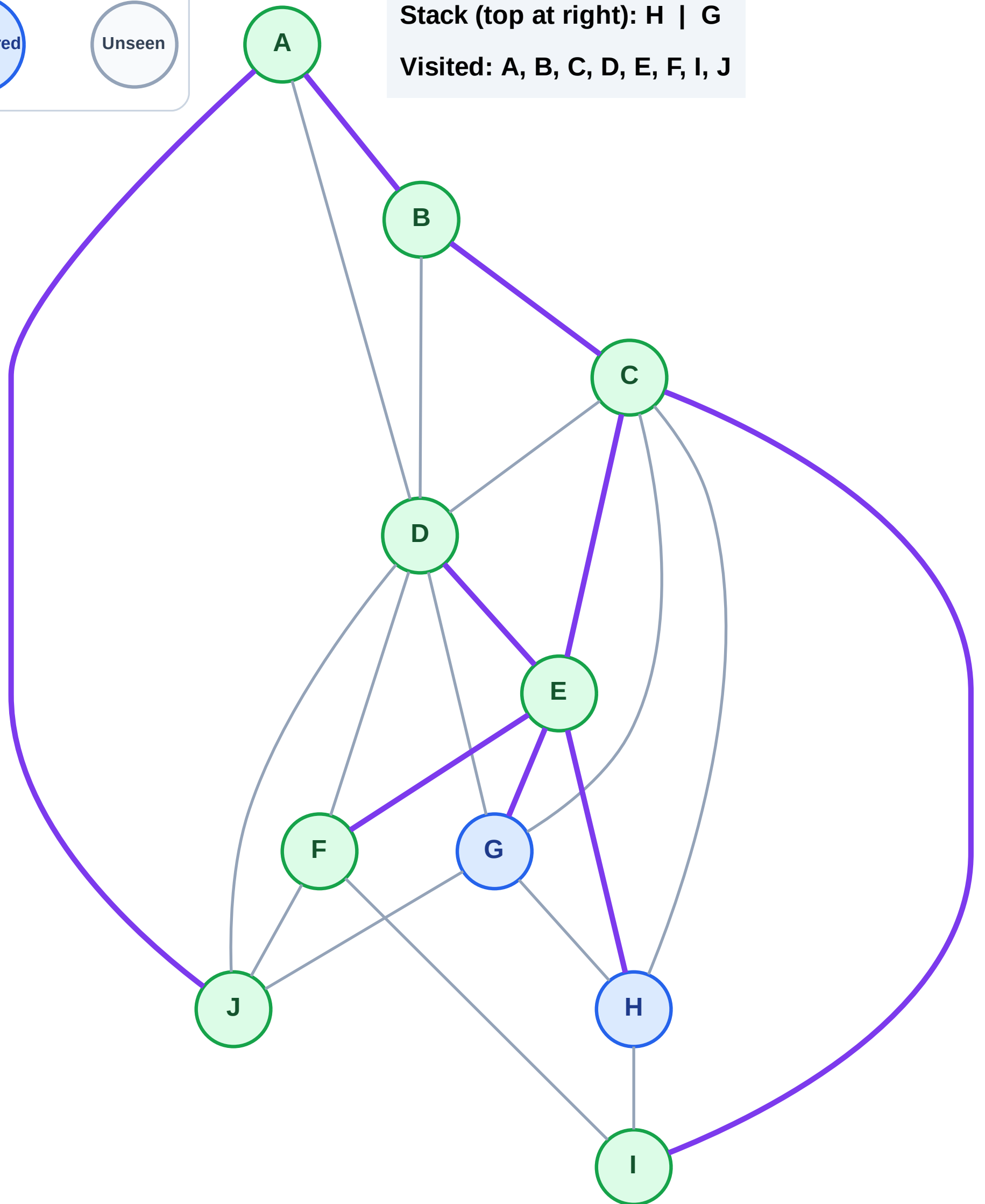
Step 17: No new neighbors from F

Legend



Stack (top at right): H | G

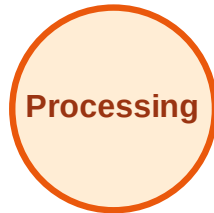
Visited: A, B, C, D, E, F, I, J



DFS Traversal

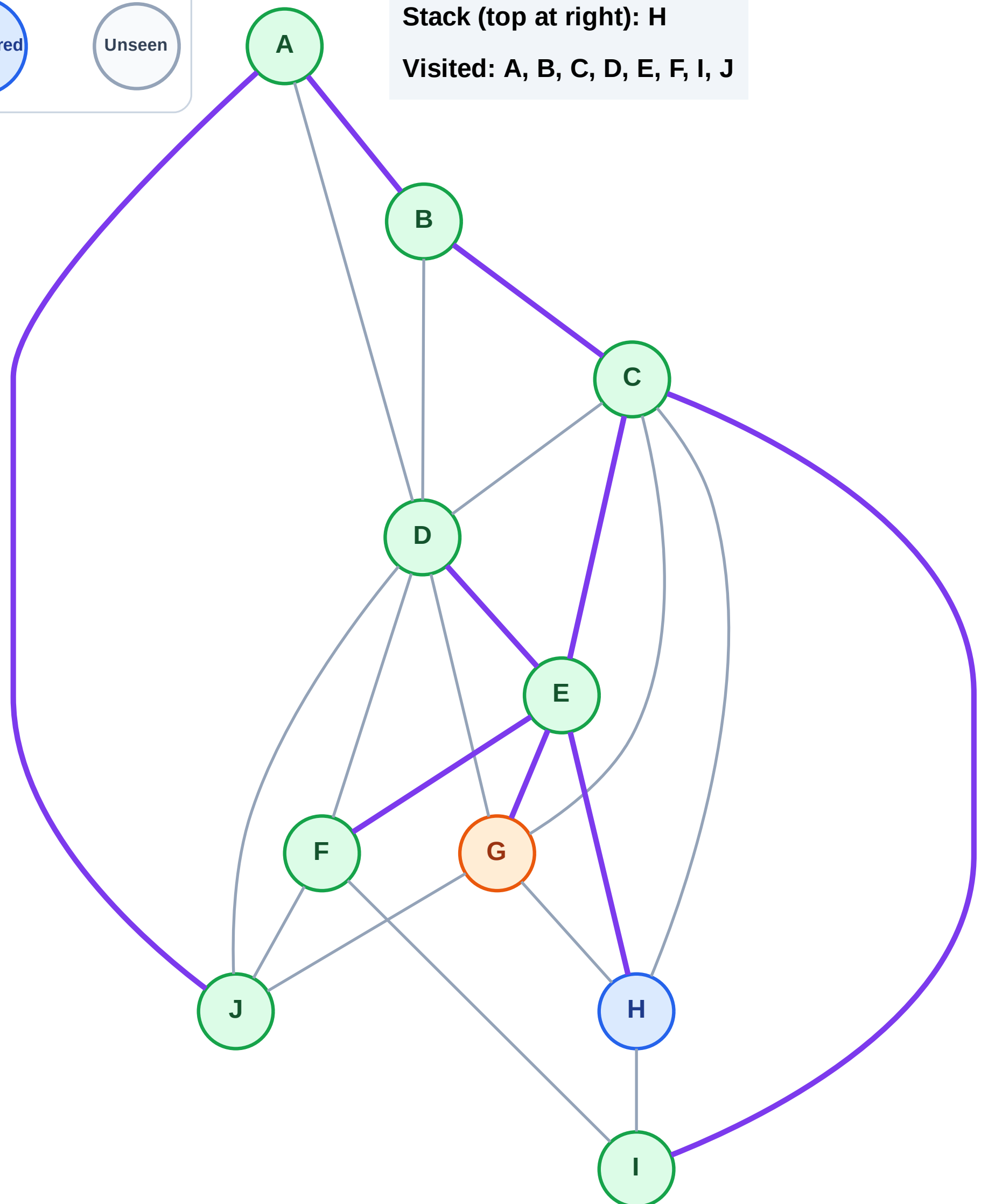
Step 18: Process node G

Legend



Stack (top at right): H

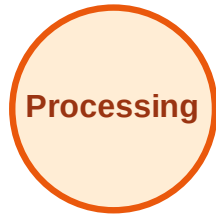
Visited: A, B, C, D, E, F, I, J



DFS Traversal

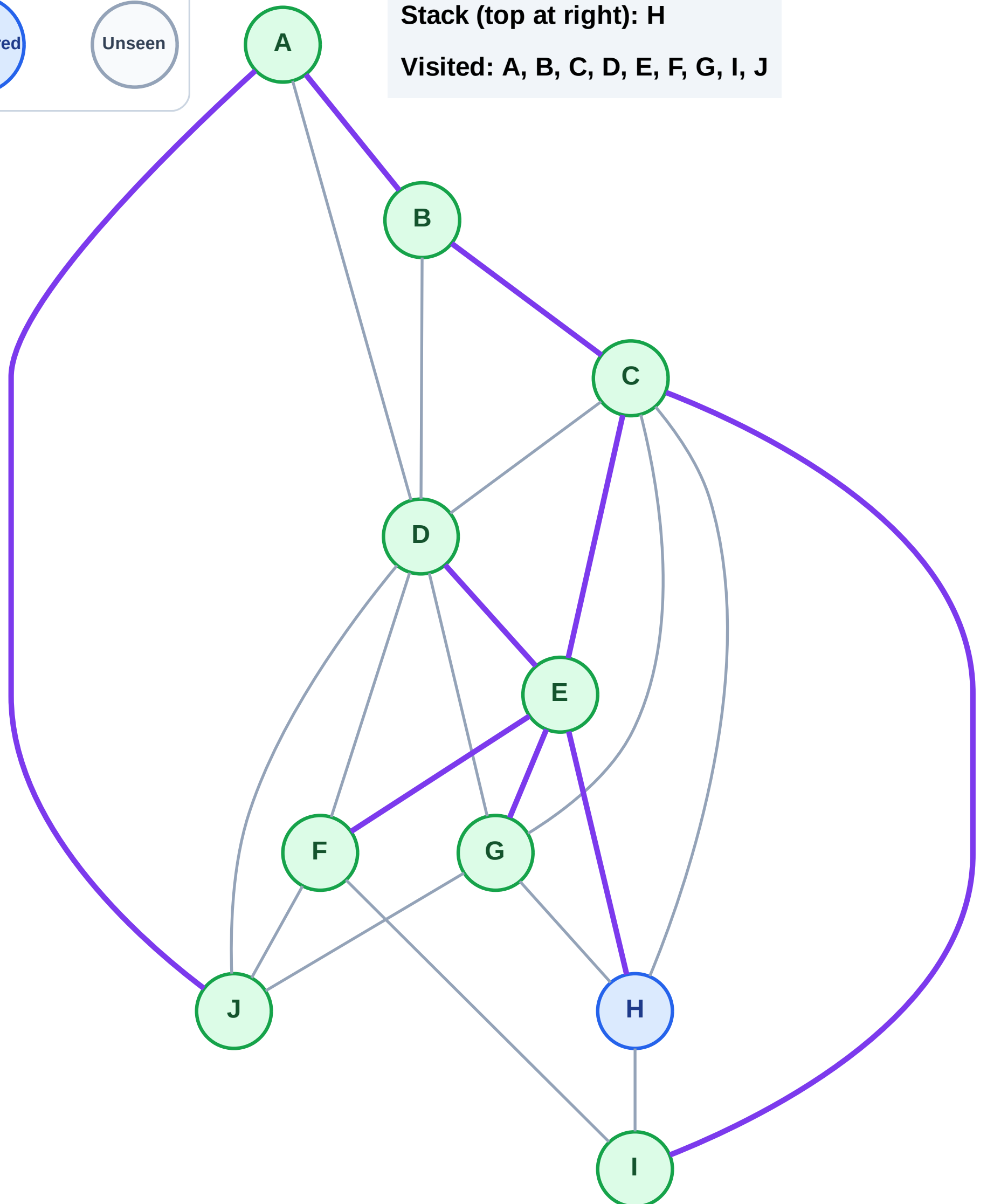
Step 19: No new neighbors from G

Legend



Stack (top at right): H

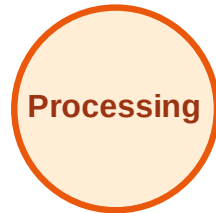
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

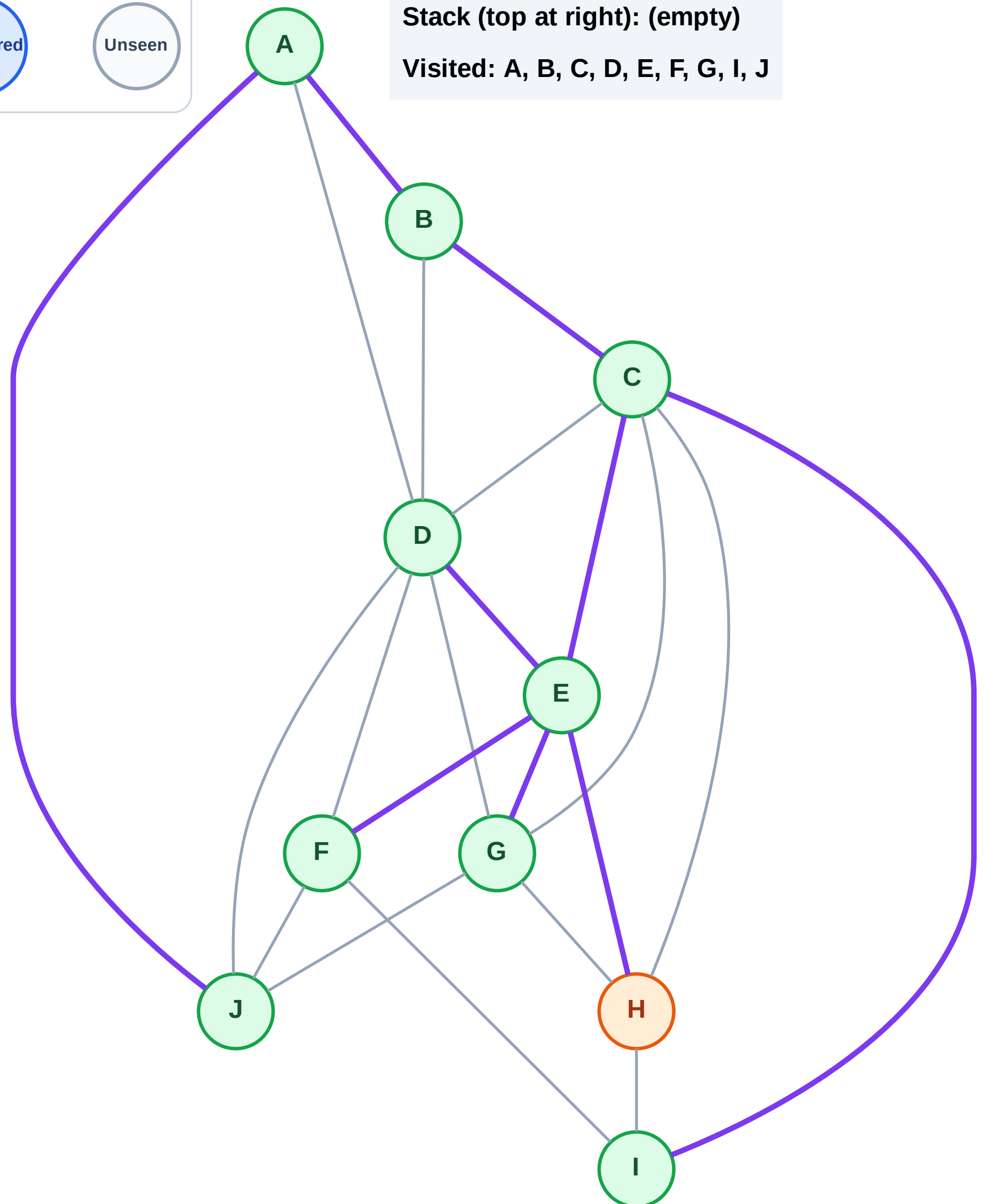
Step 20: Process node H

Legend



Stack (top at right): (empty)

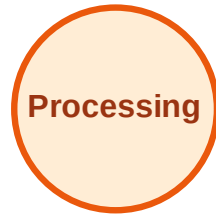
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

